

MS&E 342: Stochastic Systems and Learning Theory with Applications in Finance

Lecture 5-12: Stochastic Controls in Continuous Time

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1 Motivation and overview

Many problems in finance and modern machine learning share a common structure: an agent must make decisions over time in an uncertain and evolving environment.

In finance, typical examples include:

- allocating wealth dynamically across assets,
- deciding when to exercise an option,
- managing risk in the presence of random market movements.

In all of these problems, decisions made today affect both future opportunities and future outcomes.

A similar structure arises in machine learning. In reinforcement learning, an agent interacts with an environment, observes data sequentially, and updates its actions in order to improve long-term performance. More recent developments, such as fine-tuning large models via feedback, can also be viewed as optimizing decisions over time.

The key challenge in all these settings is the same:

How should one act today, given uncertainty about the future, in order to achieve the best long-term outcome?

Stochastic control provides a systematic framework to address this question. It combines three essential components:

- **random dynamics:** the environment evolves unpredictably,
- **sequential decisions:** actions are taken over time,
- **optimization:** the goal is to maximize a cumulative objective.

This perspective allows us to study a range of problems within a unified framework, including continuous-time models in finance, optimal stopping problems, and modern reinforcement learning methods.

The goal of this part of the course is to develop these ideas in a coherent and mathematically rigorous way. We begin with continuous-time stochastic control models and introduce the main analytical tools, including the Dynamic Programming Principle (DPP), the Hamilton–Jacobi–Bellman (HJB) equation, and the Maximum Principle. We then study optimal stopping problems and their connections to sequential decision-making. Building on these foundations, we discuss continuous-time reinforcement learning and its applications in finance, as well as modern developments such as fine-tuning generative models via feedback.

2 Introduction

2.1 The classical finite-horizon setup

We work on a filtered probability space

$$(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$$

supporting a d -dimensional Brownian motion $W = (W_t)_{0 \leq t \leq T}$. The filtration $\mathbb{F}$ represents the information available over time.

The state of the system is an $\mathbb{R}^n$ -valued process $X = (X_t)$, and the controller chooses an $\mathbb{R}^m$ -valued process $\alpha = (\alpha_t)$ with values in a control set $A \subseteq \mathbb{R}^m$. In the classical diffusion model, the state evolves according to the controlled stochastic differential equation

$$dX_s = b(s, X_s, \alpha_s) ds + \sigma(s, X_s, \alpha_s) dW_s, \quad s \in [t, T], \quad X_t = x. \quad (1)$$

Here

- b is the drift,
- σ is the volatility matrix,
- α_s is the action chosen at time s .

We assume the coefficients are such that for every admissible control there exists a unique solution to (1). The controller is rewarded through a running payoff f and a terminal payoff g . Thus the performance of a control α started from (t, x) is

$$J(t, x; \alpha) = \mathbb{E} \left[\int_t^T f(s, X_s^{t,x,\alpha}, \alpha_s) ds + g(X_T^{t,x,\alpha}) \mid X_t = x \right]. \quad (2)$$

The associated value function is

$$V(t, x) = \sup_{\alpha \in \mathcal{A}(t)} J(t, x; \alpha), \quad (3)$$

where $\mathcal{A}(t)$ denotes the set of admissible controls on $[t, T]$.

Definition 2.1 (Admissible control). *An admissible control is typically an $\mathbb{F}$ -progressively measurable, A -valued process satisfying enough integrability and state-constraint conditions so that the SDE and payoff are well-defined.*

The sign convention in (2) is for reward maximization. Cost minimization problems are entirely equivalent: one simply minimizes a cost functional or maximizes its negative.

Remark 2.2 (Infinite horizon). *On an infinite horizon one often studies*

$$\sup_{\alpha} \mathbb{E} \left[\int_0^{\infty} e^{-\rho s} f(X_s, \alpha_s) ds \right],$$

with discount factor $\rho > 0$. Many ideas remain the same, but time-homogeneous structures and stability issues become more important.

2.2 Classes of admissible controls

The notion of a “control” depends on the admissibility constraints imposed on the control process. Different modeling choices lead to different admissible classes, which in turn affect both the formulation and the analysis of the control problem.

Control type	Description
Open-loop	α is an $\mathbb{F}$ -progressively measurable process taking values in A ; it may depend on time and the realized noise, but not explicitly on the current state.
Closed-loop / history-dependent	α is $\mathbb{F}$ -adapted and may depend on the entire past trajectory $(X_s)_{0 \leq s \leq t}$.
Feedback (Markov)	$\alpha_t = a(t, X_t)$ for a measurable function a ; the control is a function of the current time and state.
Relaxed	The control is a measurable process taking values in the space of probability measures on A ; at each time, actions are randomized. This formulation is useful for compactness and existence results.
Singular / impulse	The control acts through a process of finite variation (possibly singular) or via discrete intervention times; typical in models with instantaneous adjustments or interventions.
Stopping or mixed control-stopping	The controller selects both a control process α and a stopping time τ .

An essential requirement is that admissible controls are *non-anticipative*, i.e., α_t must be measurable with respect to the information available up to time t . Controls depending on future information are called anticipating and are excluded from the classical framework.

Remark 2.3 (Feedback controls). *In Markovian settings, feedback (Markov) controls are particularly natural, as they specify a policy mapping the current state to an action. This structure is central to the dynamic programming approach.*

2.3 Value function and optimal controls

The central objects of a stochastic control problem are:

- the *value function* $V(t, x)$, defined as the supremum of the performance functional over all admissible controls,
- an *optimal control* α^* , if it exists, satisfying

$$V(t, x) = J(t, x; \alpha^*).$$

In general, explicit solutions are rarely available. Instead, one seeks characterizations of the value function and optimal controls that enable qualitative analysis, numerical approximation, or the construction of candidate optimal policies.

Two fundamental approaches underpin the theory:

1. **Dynamic programming:** the optimization problem is decomposed over time, leading to the dynamic programming principle and, in the Markovian case, to the Hamilton–Jacobi–Bellman (HJB) equation.

2. **Variational analysis:** optimality conditions are derived by perturbing the control along admissible directions, leading to Pontryagin's maximum principle.

These approaches provide complementary characterizations of optimal controls and value functions.

2.4 The dynamic programming principle

The dynamic programming principle is Bellman's principle of optimality in continuous time. Informally:

If a strategy is optimal from today onward, then after any intermediate time, the continuation of that strategy must still be optimal from the random state reached at that time.

Proposition 2.4 (DPP, informal statement). *Let τ be any stopping time with values in $[t, T]$. Then*

$$V(t, x) = \sup_{\alpha \in \mathcal{A}(t)} \mathbb{E} \left[\int_t^\tau f(s, X_s^{t,x,\alpha}, \alpha_s) ds + V(\tau, X_\tau^{t,x,\alpha}) \right]. \quad (4)$$

The meaning of (4) is worth pausing over. The value function compresses the entire future into a single object. Once we know V , the continuation problem from time τ onward is already solved. This is why dynamic programming is so powerful: it turns an optimization over all future decisions into a recursive equation.

A proof of the DPP uses two simple but profound ideas:

- split the payoff into the part earned before τ and the part after τ ,
- condition on the information available at time τ and use the tower property of conditional expectation.

In many texts, the DPP is the conceptual heart of the subject. The HJB equation is its infinitesimal shadow.

2.5 From DPP to the HJB equation

Suppose now that the problem is Markovian and the value function is sufficiently smooth. For a smooth test function φ , define the controlled second-order operator

$$\mathcal{L}^a \varphi(t, x) = b(t, x, a) \cdot \nabla_x \varphi(t, x) + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, a) \nabla_x^2 \varphi(t, x)).$$

If we apply the DPP over a very short time interval $[t, t+h]$, use Itô's formula, and divide by h , we are led formally to the *Hamilton–Jacobi–Bellman equation*

$$\partial_t V(t, x) + \sup_{a \in A} \left\{ \mathcal{L}^a V(t, x) + f(t, x, a) \right\} = 0, \quad (t, x) \in [0, T] \times \mathbb{R}^n, \quad (5)$$

with terminal condition

$$V(T, x) = g(x).$$

Equation (5) is Bellman's equation in differential form. It says that, at each point (t, x) , one chooses the control a that makes the local rate of increase of value as large as possible.

Remark 2.5 (The verification idea). *A classical strategy is:*

1. *guess a smooth solution v to the HJB equation,*
2. *identify a maximizer $\hat{a}(t, x)$ in the Hamiltonian,*
3. *prove that $v = V$ and that the feedback rule $\alpha_t^* = \hat{a}(t, X_t)$ is optimal.*

This is called a verification theorem. The argument itself does not require uniqueness of HJB solutions in advance; one identifies the candidate directly with the value function. Uniqueness becomes central later, especially in the viscosity framework, through comparison principles.

Remark 2.6 (Why future lectures need viscosity solutions). *The value function need not be smooth. Corners, constraints, stopping features, and degenerate volatility can all destroy classical differentiability. The right weak notion of solution for HJB equations is often the viscosity solution. This is one of the major achievements of modern stochastic control, but for now the main message is simple: the HJB equation remains correct, even when smooth calculus is no longer available in the classical sense.*

2.6 Pontryagin's maximum principle: the trajectory viewpoint

Dynamic programming characterizes the value of being at a point (t, x) through a global object, the value function. Pontryagin's maximum principle takes a different perspective: it provides *first-order necessary conditions that must hold along an optimal trajectory*.

If a control is optimal, then any infinitesimal perturbation of it should not improve the objective; what condition does this impose along the resulting trajectory?

This leads to a local, variational characterization of optimality. In contrast to dynamic programming, the maximum principle does not aim to construct the value function on the entire state space. Rather, it yields a system of equations that directly characterizes the *optimal control and trajectory*. This viewpoint is particularly natural when the state is not Markovian, when one seeks first-order conditions, or when the state space is too large for PDE methods to be practical.

For a controlled diffusion, define the Hamiltonian

$$H(t, x, a, y, z) = f(t, x, a) + b(t, x, a) \cdot y + \text{Tr}(\sigma(t, x, a)^\top z), \quad (6)$$

where $y \in \mathbb{R}^n$ and $z \in \mathbb{R}^{n \times d}$.

Suppose (X^*, α^*) is an optimal state-control pair. Then there exists an *adjoint process* (Y, Z) solving the backward stochastic differential equation

$$dY_t = -\nabla_x H(t, X_t^*, \alpha_t^*, Y_t, Z_t) dt + Z_t dW_t, \quad Y_T = \nabla g(X_T^*), \quad (7)$$

such that the control satisfies the pointwise maximization condition

$$\alpha_t^* \in \arg \max_{a \in A} H(t, X_t^*, a, Y_t, Z_t) \quad \text{for a.e. } t, \mathbb{P}\text{-a.s.} \quad (8)$$

The content of the maximum principle is therefore the following: the optimal control is obtained by maximizing the Hamiltonian *pointwise in time*, once the adjoint process is known. Consequently, the original control problem is reduced to solving a *coupled forward-backward system* consisting of

- the forward SDE for X^* ,
- the backward SDE for (Y, Z) ,
- and the pointwise maximization condition (8).

In many models, the maximization condition yields an explicit formula for α^* in terms of (X^*, Y, Z) . Under additional convexity or concavity assumptions, these necessary conditions are also sufficient.

Remark 2.7 (HJB versus maximum principle). *The two viewpoints are complementary but conceptually distinct:*

- *The HJB equation is a global characterization of the value function on the state space.*
- *The maximum principle is a local characterization of optimality along a single trajectory.*

In general, the maximum principle does not directly yield the value function. However, in the smooth Markovian setting, one often has

$$Y_t = \nabla_x V(t, X_t^*),$$

so that the Hamiltonian maximization condition coincides with the optimizer appearing in the HJB equation. In this sense, the adjoint process plays the role of the gradient of the value function along the optimal path.

2.7 Example: Merton's portfolio problem

A canonical introductory example is Merton's optimal investment problem. It captures the essential structure of a stochastic control problem while remaining sufficiently tractable to admit an explicit solution.

Model. There are two assets:

- a bank account S_t^0 satisfying

$$dS_t^0 = rS_t^0 dt,$$

where r is the constant interest rate;

- a risky stock S_t satisfying the Black–Scholes model

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

with constants $\mu > r$ and $\sigma > 0$.

Let π_t denote the fraction of wealth invested in the stock at time t . Then the wealth process X satisfies

$$dX_t = X_t(r + \pi_t(\mu - r)) dt + X_t \pi_t \sigma dW_t, \quad X_0 = x > 0. \quad (9)$$

We choose the portfolio process $\pi = (\pi_t)$ to maximize expected utility of terminal wealth:

$$V(t, x) = \sup_{\pi \in \mathcal{A}(t)} \mathbb{E}[U(X_T^{t,x,\pi})]. \quad (10)$$

Assume CRRA utility

$$U(x) = \frac{x^p}{p}, \quad p < 1, \quad p \neq 0.$$

(When $0 < p < 1$, this is the most common textbook case.)

The HJB equation. Since there is no running reward, the HJB equation becomes

$$\partial_t V(t, x) + \sup_{a \in A} \left\{ x(r + a(\mu - r)) V_x(t, x) + \frac{1}{2} \sigma^2 a^2 x^2 V_{xx}(t, x) \right\} = 0, \quad (11)$$

with terminal condition $V(T, x) = U(x)$.

The homogeneity of CRRA utility suggests the ansatz

$$V(t, x) = \phi(t) \frac{x^p}{p}.$$

Then

$$V_x(t, x) = \phi(t) x^{p-1}, \quad V_{xx}(t, x) = \phi(t) (p-1) x^{p-2}.$$

Substituting into (11) gives

$$\frac{\phi'(t)}{p} + \phi(t) \sup_{a \in A} \left\{ r + a(\mu - r) + \frac{1}{2} (p-1) \sigma^2 a^2 \right\} = 0.$$

Equivalently,

$$\phi'(t) + p \phi(t) \sup_{a \in A} \left\{ r + a(\mu - r) + \frac{1}{2} (p-1) \sigma^2 a^2 \right\} = 0, \quad \phi(T) = 1. \quad (12)$$

Optimal portfolio. If $A = \mathbb{R}$, the maximization is over a concave quadratic function of a , and the first-order condition yields

$$\pi^* = \frac{\mu - r}{(1-p)\sigma^2}. \quad (13)$$

This is the celebrated Merton proportion: invest a constant fraction of wealth in the risky asset. Higher risk premia $(\mu - r)$ increase the risky allocation, while larger volatility σ and larger risk aversion $(1-p)$ reduce it.

Plugging (13) into (12) gives

$$\phi'(t) + p \left(r + \frac{(\mu - r)^2}{2(1-p)\sigma^2} \right) \phi(t) = 0, \quad \phi(T) = 1,$$

so

$$\phi(t) = \exp \left(p \left(r + \frac{(\mu - r)^2}{2(1-p)\sigma^2} \right) (T - t) \right). \quad (14)$$

Therefore

$$V(t, x) = \exp \left(p \left(r + \frac{(\mu - r)^2}{2(1-p)\sigma^2} \right) (T - t) \right) \frac{x^p}{p}. \quad (15)$$

Remark 2.8 (Insights from Merton's problem). *Merton's optimal investment problem exhibits several key structural features:*

- *The optimal policy is homothetic: the investor allocates a constant fraction of wealth to the risky asset, so the strategy scales linearly with wealth.*

- *The optimal allocation*

$$\pi^* = \frac{\mu - r}{(1 - \gamma)\sigma^2}$$

separates market parameters and preferences: it increases with the risk premium $\mu - r$, and decreases with volatility σ and risk aversion.

- *The tractability of the problem relies on the homogeneity of CRRA utility, which reduces the HJB equation to an ordinary differential equation via the ansatz $V(t, x) = \phi(t)x^\gamma/\gamma$.*
- *This example illustrates the general stochastic control workflow: formulation of the value function, derivation of the HJB equation, pointwise optimization, and recovery of an optimal feedback control.*

2.8 Example: a linear–quadratic control problem

We now consider a canonical class of stochastic control problems where both the dynamics and the objective are quadratic. These problems admit explicit solutions and illustrate how the HJB equation reduces to a system of ordinary differential equations.

Model. Fix $q, g \geq 0$ and $\sigma > 0$. The state process satisfies

$$dX_s = a_s ds + \sigma dW_s, \quad s \in [t, T], \quad X_t = x, \quad (16)$$

where the control $a = (a_s)$ is an $\mathbb{F}$ -progressively measurable process with values in $\mathbb{R}$.

The objective is to minimize the quadratic cost

$$J(t, x; a) = \mathbb{E} \left[\int_t^T \left(\frac{1}{2} a_s^2 + \frac{q}{2} X_s^2 \right) ds + \frac{g}{2} X_T^2 \right]. \quad (17)$$

The value function is

$$V(t, x) = \inf_{a \in \mathcal{A}(t)} J(t, x; a). \quad (18)$$

The HJB equation. The HJB equation associated with (18) is

$$\partial_t V(t, x) + \inf_{a \in \mathbb{R}} \left\{ a V_x(t, x) + \frac{1}{2} \sigma^2 V_{xx}(t, x) + \frac{1}{2} a^2 + \frac{q}{2} x^2 \right\} = 0, \quad (19)$$

with terminal condition

$$V(T, x) = \frac{g}{2} x^2.$$

Quadratic ansatz. The quadratic structure of the problem suggests that the value function is quadratic in the state. We therefore consider the ansatz

$$V(t, x) = \frac{1}{2} P(t) x^2 + r(t), \quad (20)$$

for deterministic functions P and r .

Then

$$V_x(t, x) = P(t)x, \quad V_{xx}(t, x) = P(t).$$

Pointwise minimization. Substituting into (19), the minimization over a becomes

$$\inf_{a \in \mathbb{R}} \left\{ a P(t)x + \frac{1}{2}a^2 \right\}.$$

This is a strictly convex quadratic function of a , and the first-order condition yields

$$a^*(t, x) = -P(t)x. \quad (21)$$

The minimal value is

$$-\frac{1}{2}P(t)^2x^2.$$

Riccati equation. Substituting the optimal value back into the HJB equation, we obtain

$$\frac{1}{2}P'(t)x^2 + r'(t) - \frac{1}{2}P(t)^2x^2 + \frac{1}{2}\sigma^2P(t) + \frac{q}{2}x^2 = 0.$$

Matching coefficients in x^2 and constants yields the system

$$\begin{cases} P'(t) - P(t)^2 + q = 0, & P(T) = g, \\ r'(t) + \frac{1}{2}\sigma^2P(t) = 0, & r(T) = 0. \end{cases} \quad (22)$$

The first equation is a *Riccati differential equation*. Once P is determined, the function r is obtained by integration.

Optimal control and closed-loop dynamics. The optimal control is the linear feedback law

$$a_s^* = -P(s)X_s^*. \quad (23)$$

Substituting into (16), the optimal state process satisfies the closed-loop SDE

$$dX_s^* = -P(s)X_s^* ds + \sigma dW_s, \quad X_t^* = x.$$

Remark 2.9 (Structure of linear-quadratic control). *This example highlights several key features of stochastic control:*

- *The quadratic structure of the cost and linear dynamics lead to a quadratic value function and a linear feedback control.*
- *The HJB equation reduces to a finite-dimensional system of ODEs, namely the Riccati equation (22).*
- *The optimal control is state-dependent and stabilizing: when X_s is large, the control acts more strongly to push it toward zero.*
- *Linear-quadratic problems form a fundamental class in stochastic control and appear in a wide range of applications, including optimal execution, inventory control, and continuous-time reinforcement learning.*

3 Hamilton–Jacobi–Bellman equations

The Hamilton–Jacobi–Bellman equation is the central analytical object of the dynamic programming approach. The discussion in the introduction explained informally how the HJB equation arises from the dynamic programming principle. We now isolate this mechanism in a more systematic way, state the classical verification argument, and explain why viscosity solutions are indispensable once smoothness is lost.

Throughout this section we remain in the finite-horizon Markovian diffusion setting (1)–(3).

3.1 The Bellman operator

For each fixed control value $a \in A$, recall the second-order differential operator

$$\mathcal{L}^a \varphi(t, x) = b(t, x, a) \cdot \nabla_x \varphi(t, x) + \frac{1}{2} \operatorname{Tr}(\sigma \sigma^\top(t, x, a) \nabla_x^2 \varphi(t, x)).$$

The dynamic programming principle suggests introducing the *Bellman operator*

$$\mathcal{H}(t, x, p, M) = \sup_{a \in A} \left\{ b(t, x, a) \cdot p + \frac{1}{2} \operatorname{Tr}(\sigma \sigma^\top(t, x, a) M) + f(t, x, a) \right\}, \quad (24)$$

where $p \in \mathbb{R}^n$ and $M \in \mathbb{R}^{n \times n}$ is symmetric.

In this notation, the HJB equation takes the compact form

$$\partial_t V(t, x) + \mathcal{H}(t, x, \nabla_x V(t, x), \nabla_x^2 V(t, x)) = 0, \quad (t, x) \in [0, T] \times \mathbb{R}^n, \quad (25)$$

with terminal condition

$$V(T, x) = g(x). \quad (26)$$

Several structural features are already visible in (25). First, the equation is nonlinear because the optimization over actions is carried out *inside* the PDE. Second, when the control enters the diffusion coefficient σ , the equation is typically *fully nonlinear*; if the control enters only through the drift, the equation is still nonlinear, but often only semilinear. Third, the terminal condition (26) reflects the fact that at time T no further decisions remain, so the continuation value reduces exactly to the terminal reward.

3.2 Dynamic programming principle

To keep the main mechanism transparent, let us first state the dynamic programming principle for a deterministic intermediate time $r \in [t, T]$. The stopping-time version from the introduction is obtained by the same decomposition and pasting argument, with additional measurability bookkeeping.

Proposition 3.1 (DPP for a deterministic intermediate time). *Fix $t \leq r \leq T$. Assume admissible controls are stable under concatenation at time r , and that for every $\varepsilon > 0$ one may choose measurable ε -optimal continuation controls for the problem started from (r, y) . Then*

$$V(t, x) = \sup_{\alpha \in \mathcal{A}(t)} \mathbb{E} \left[\int_t^r f(s, X_s^{t,x,\alpha}, \alpha_s) ds + V(r, X_r^{t,x,\alpha}) \right]. \quad (27)$$

Proof. We prove the two inequalities separately.

The “ $\leq$ ” direction. Fix an admissible control $\alpha \in \mathcal{A}(t)$ and write $X = X^{t,x,\alpha}$. Splitting the payoff at time r gives

$$J(t, x; \alpha) = \mathbb{E} \left[\int_t^r f(s, X_s, \alpha_s) ds + J(r, X_r; \alpha^r) \right],$$

where α^r denotes the continuation of α after time r . Since $J(r, X_r; \alpha^r) \leq V(r, X_r)$ by definition of the value function, we obtain

$$J(t, x; \alpha) \leq \mathbb{E} \left[\int_t^r f(s, X_s, \alpha_s) ds + V(r, X_r) \right].$$

Taking the supremum over α yields

$$V(t, x) \leq \sup_{\alpha \in \mathcal{A}(t)} \mathbb{E} \left[\int_t^r f(s, X_s^{t,x,\alpha}, \alpha_s) ds + V(r, X_r^{t,x,\alpha}) \right].$$

The “ $\geq$ ” direction. Fix $\varepsilon > 0$ and an admissible control segment on $[t, r]$, still denoted by α . For each possible state y at time r , choose an ε -optimal continuation control $\beta^{\varepsilon, y} \in \mathcal{A}(r)$ such that

$$J(r, y; \beta^{\varepsilon, y}) \geq V(r, y) - \varepsilon.$$

By the measurable selection and concatenation assumptions, we may paste these controls to form an admissible control $\tilde{\alpha}^\varepsilon$ that follows α up to time r and then switches to β^{ε, X_r} . For this pasted control,

$$J(t, x; \tilde{\alpha}^\varepsilon) = \mathbb{E} \left[\int_t^r f(s, X_s, \alpha_s) ds + J(r, X_r; \beta^{\varepsilon, X_r}) \right] \geq \mathbb{E} \left[\int_t^r f(s, X_s, \alpha_s) ds + V(r, X_r) \right] - \varepsilon.$$

Since $V(t, x) \geq J(t, x; \tilde{\alpha}^\varepsilon)$, taking the supremum over the initial segment α and then letting $\varepsilon \downarrow 0$ gives the reverse inequality. $\square$

Remark 3.2 (Where the subtlety lies). *The first inequality is essentially just conditioning and the definition of V . The second is the genuinely delicate part: one must be able to choose nearly optimal continuation controls in a measurable way and paste them back into the original control. This is the technical core of the DPP in continuous time.*

The stopping-time version stated in the introduction is obtained by replacing r with a stopping time τ and conditioning on $\mathcal{F}_\tau$. No new idea appears, but the measurability of the continuation control must then be handled more carefully.

3.3 From DPP to HJB equation

Let $h > 0$ be small and set $\tau = t + h$. The dynamic programming principle suggests that

$$V(t, x) = \sup_{\alpha \in \mathcal{A}(t)} \mathbb{E} \left[\int_t^{t+h} f(s, X_s^{t,x,\alpha}, \alpha_s) ds + V(t+h, X_{t+h}^{t,x,\alpha}) \right]. \quad (28)$$

Assume for the moment that $V \in C^{1,2}([0, T] \times \mathbb{R}^n)$ and that all expectations below are justified.

Fix a control value $a \in A$ and consider using this action over the short interval $[t, t + h]$. If X^a denotes the corresponding state process, Itô's formula gives

$$V(t + h, X_{t+h}^a) = V(t, x) + \int_t^{t+h} \left(\partial_s V + \mathcal{L}^a V \right)(s, X_s^a) ds + \int_t^{t+h} \nabla_x V(s, X_s^a)^\top \sigma(s, X_s^a, a) dW_s.$$

Taking expectations eliminates the martingale term and yields

$$\mathbb{E}[V(t + h, X_{t+h}^a)] = V(t, x) + \mathbb{E} \int_t^{t+h} \left(\partial_s V + \mathcal{L}^a V \right)(s, X_s^a) ds.$$

Substituting this into (28), subtracting $V(t, x)$ from both sides, and dividing by h , we are led formally to

$$0 = \sup_{a \in A} \left\{ \partial_t V(t, x) + \mathcal{L}^a V(t, x) + f(t, x, a) \right\},$$

which is precisely (25).

This short-time argument is the infinitesimal form of dynamic programming. The dynamic programming principle is a global recursive statement over time intervals; the HJB equation is its differential counterpart obtained by zooming in on an interval of length h and letting $h \downarrow 0$.

Remark 3.3 (Why this derivation is only informal). *The preceding computation assumes smoothness of V and enough regularity to justify Itô's formula, interchange of limits, and evaluation at the point (t, x) . In many important models the value function is not $C^{1,2}$, even when the coefficients are smooth. The dynamic programming principle remains valid, however, and the correct rigorous interpretation of the HJB equation is then given by viscosity solutions.*

3.4 Interpretation of the equation

The HJB equation may be read as a pointwise balance of instantaneous effects. At a state (t, x) , the quantity

$$\partial_t V(t, x) + \mathcal{L}^a V(t, x) + f(t, x, a)$$

represents the local rate of change of the total value when action a is used. The controller then chooses the action that maximizes this local rate.

Each term has a clear interpretation:

- $\partial_t V$ measures how the value changes as the remaining horizon shrinks;
- $b \cdot \nabla_x V$ is the contribution of the controlled drift to the continuation value;
- $\frac{1}{2} \text{Tr}(\sigma \sigma^\top \nabla_x^2 V)$ captures the effect of uncertainty, and depends on the curvature of the value function;
- f is the instantaneous reward earned by exercising action a at the current state.

The second-order term deserves special emphasis. In deterministic control, only first derivatives appear. In stochastic control, volatility interacts with the curvature of the value function, so uncertainty affects value at second order. This is one of the most important conceptual differences between deterministic and stochastic optimization.

3.5 A classical verification theorem

The essential content of verification is simple. A smooth solution of the HJB equation dominates the payoff generated by every admissible control. Hence the only remaining step is to find one admissible control for which this inequality becomes an equality along the corresponding trajectory.

Proposition 3.4 (Verification theorem). *Let $v \in C^{1,2}([0, T] \times \mathbb{R}^n) \cap C([0, T] \times \mathbb{R}^n)$ satisfy*

$$\partial_t v(t, x) + \sup_{a \in A} \left\{ \mathcal{L}^a v(t, x) + f(t, x, a) \right\} = 0, \quad v(T, x) = g(x).$$

Assume that for every admissible control α , Itô's formula may be applied to $v(s, X_s^{t,x,\alpha})$ and that the resulting stochastic integral is a true martingale. If there exists an admissible control α^ with associated state process $X^* = X^{t,x,\alpha^*}$ such that*

$$\partial_s v(s, X_s^*) + \mathcal{L}^{\alpha_s^*} v(s, X_s^*) + f(s, X_s^*, \alpha_s^*) = 0 \quad \text{for a.e. } s \in [t, T], \quad \mathbb{P}\text{-a.s.}, \quad (29)$$

then

$$v(t, x) = V(t, x) = J(t, x; \alpha^*).$$

Proof. Fix an admissible control α and write $X_s = X_s^{t,x,\alpha}$, $s \in [t, T]$.

Since X satisfies

$$dX_s = b_s ds + \sigma_s dW_s,$$

and since $v \in C^{1,2}$, Itô's formula gives

$$\begin{aligned} dv(s, X_s) &= \partial_s v(s, X_s) ds + \nabla_x v(s, X_s)^\top dX_s + \frac{1}{2} \text{Tr} \left(d\langle X \rangle_s \nabla_x^2 v(s, X_s) \right) \\ &= \left[\partial_s v(s, X_s) + b(s, X_s, \alpha_s) \cdot \nabla_x v(s, X_s) + \frac{1}{2} \text{Tr} \left(\sigma \sigma^\top(s, X_s, \alpha_s) \nabla_x^2 v(s, X_s) \right) \right] ds \\ &\quad + \nabla_x v(s, X_s)^\top \sigma(s, X_s, \alpha_s) dW_s \\ &= [\partial_s v(s, X_s) + \mathcal{L}^{\alpha_s} v(s, X_s)] ds + \nabla_x v(s, X_s)^\top \sigma(s, X_s, \alpha_s) dW_s \end{aligned}$$

where the last equation holds by the definition of the generator $\mathcal{L}^a$. Integrating from t to T yields

$$v(T, X_T) = v(t, x) + \int_t^T [\partial_s v(s, X_s) + \mathcal{L}^{\alpha_s} v(s, X_s)] ds + \int_t^T \nabla_x v(s, X_s)^\top \sigma(s, X_s, \alpha_s) dW_s.$$

The last term is a stochastic integral. By the martingale assumption in the statement of the theorem,

$$\mathbb{E} \left[\int_t^T \nabla_x v(s, X_s)^\top \sigma(s, X_s, \alpha_s) dW_s \right] = 0.$$

Using also the terminal condition $v(T, \cdot) = g(\cdot)$, we obtain

$$\mathbb{E}[g(X_T)] = v(t, x) + \mathbb{E} \int_t^T [\partial_s v(s, X_s) + \mathcal{L}^{\alpha_s} v(s, X_s)] ds.$$

Therefore

$$\begin{aligned} J(t, x; \alpha) &= \mathbb{E} \left[\int_t^T f(s, X_s, \alpha_s) ds + g(X_T) \right] \\ &= v(t, x) + \mathbb{E} \int_t^T [\partial_s v(s, X_s) + \mathcal{L}^{\alpha_s} v(s, X_s) + f(s, X_s, \alpha_s)] ds. \end{aligned}$$

Now the HJB equation says

$$\partial_t v(t, x) + \sup_{a \in A} \{ \mathcal{L}^a v(t, x) + f(t, x, a) \} = 0.$$

Hence, for every fixed $a \in A$,

$$\partial_t v(t, x) + \mathcal{L}^a v(t, x) + f(t, x, a) \leq 0.$$

Consequently,

$$J(t, x; \alpha) \leq v(t, x).$$

Since α was arbitrary, taking the supremum over all admissible controls gives

$$V(t, x) \leq v(t, x).$$

It remains to prove the reverse inequality for the candidate optimal control. Suppose that there exists an admissible control α^* with associated state process

$$X_s^* = X_s^{t, x, \alpha^*}$$

such that

$$\partial_s v(s, X_s^*) + \mathcal{L}^{\alpha_s^*} v(s, X_s^*) + f(s, X_s^*, \alpha_s^*) = 0$$

for almost every $s \in [t, T]$, almost surely. Applying the previous Itô formula calculation to α^* , we now get equality:

$$J(t, x; \alpha^*) = v(t, x) + \mathbb{E} \int_t^T \left[\partial_s v(s, X_s^*) + \mathcal{L}^{\alpha_s^*} v(s, X_s^*) + f(s, X_s^*, \alpha_s^*) \right] ds = v(t, x).$$

Since $J(t, x; \alpha^*) \leq V(t, x)$ by definition of the value function, and since we have already shown $V(t, x) \leq v(t, x)$, it follows that

$$v(t, x) = V(t, x) = J(t, x; \alpha^*).$$

Thus α^* is optimal and v is the value function. □

The proposition captures the practical use of HJB theory. One guesses or computes a candidate function v , checks the HJB equation, and then looks for a control that realizes equality along its own trajectory. This strategy is what makes explicit solutions possible in examples such as Merton's portfolio problem and linear-quadratic control in Section 2.8.

3.6 When smoothness fails: viscosity solutions

The verification theorem is elegant, but its hypotheses are strong. Even in simple models the value function may fail to be differentiable. There are several reasons for this:

- the optimizer may switch abruptly from one region of the state space to another;
- state constraints or stopping features may create kinks or free boundaries;
- the diffusion matrix may be degenerate, so the PDE is not uniformly parabolic.

The modern remedy is to interpret the HJB equation in the viscosity sense.

Definition 3.5 (Viscosity subsolution and supersolution). *Let $u : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous.*

- *We say that u is a viscosity subsolution of (25) if, whenever $\phi \in C^{1,2}([0, T] \times \mathbb{R}^n)$ and $u - \phi$ attains a local maximum at $(t_0, x_0) \in [0, T] \times \mathbb{R}^n$, one has*

$$\partial_t \phi(t_0, x_0) + \mathcal{H}(t_0, x_0, \nabla_x \phi(t_0, x_0), \nabla_x^2 \phi(t_0, x_0)) \leq 0.$$

- *We say that u is a viscosity supersolution of (25) if, whenever $\phi \in C^{1,2}([0, T] \times \mathbb{R}^n)$ and $u - \phi$ attains a local minimum at (t_0, x_0) , one has*

$$\partial_t \phi(t_0, x_0) + \mathcal{H}(t_0, x_0, \nabla_x \phi(t_0, x_0), \nabla_x^2 \phi(t_0, x_0)) \geq 0.$$

If both conditions hold and the terminal condition (26) is satisfied, then u is called a viscosity solution.

The key idea is subtle but powerful: one does not differentiate the value function directly. Instead, one compares it locally with smooth test functions that touch it from above or below. This notion is stable under approximation, compatible with dynamic programming, and robust enough to handle degenerate and nonsmooth problems.

Proposition 3.6 (Comparison principle, informal statement). *Suppose the coefficients of the Bellman operator are continuous and satisfy the standard structural conditions ensuring well-posedness of the HJB equation (for example, appropriate continuity, growth, and ellipticity assumptions, together with a prescribed growth class for solutions). Let $u, w : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous functions such that*

- *u is a viscosity subsolution of*

$$\partial_t v(t, x) + \mathcal{H}(t, x, \nabla_x v(t, x), \nabla_x^2 v(t, x)) = 0,$$

- *w is a viscosity supersolution of the same equation,*
- *and*

$$u(T, x) \leq w(T, x) \quad \text{for all } x \in \mathbb{R}^n.$$

Then

$$u(t, x) \leq w(t, x), \quad (t, x) \in [0, T] \times \mathbb{R}^n.$$

In particular, there is at most one viscosity solution in the corresponding class of continuous functions satisfying the prescribed terminal condition and growth condition.

Remark 3.7 (Why comparison matters). *The comparison principle is the key uniqueness tool in the viscosity theory of HJB equations. Once one proves that the value function V is a viscosity solution of the HJB equation, comparison implies that V is the unique viscosity solution in the relevant function class. Thus, even when classical smoothness fails and the verification theorem is no longer directly applicable, the HJB equation still characterizes the value function uniquely.*

Remark 3.8 (The modern HJB paradigm). *Under standard continuity, growth, and compactness assumptions, one typically proves the theory in two steps:*

1. the value function V is a viscosity solution of the HJB equation, obtained directly from the dynamic programming principle;
2. a comparison principle implies uniqueness in the class of viscosity solutions with the prescribed growth condition.

Together, these two statements identify the value function as the correct solution of the HJB equation, even when no classical derivatives exist.

3.7 Why the HJB equation matters

The HJB equation is much more than a formal byproduct of dynamic programming. It is the bridge between stochastic optimization and analysis. Once the control problem is written as a PDE, one gains access to structural tools from partial differential equations, convex analysis, and numerical approximation. In favorable models the HJB equation yields explicit solutions; in more complex models it provides the right object for approximation schemes, stability analysis, and eventually for reinforcement learning algorithms that aim to learn value functions and optimal policies from data.

For this reason, the HJB equation is best viewed not merely as an equation to be solved, but as a precise expression of Bellman's principle in differential form.

3.8 Examples

Example 3.9 (Finite-horizon dividend-rate control). *Consider a firm with cash reserve process X satisfying*

$$dX_s = (\mu - d_s) ds + \sigma dW_s, \quad X_t = x,$$

where $\mu > 0$ is the firm's cash-flow rate, $\sigma > 0$ is cash-flow volatility, and $d_s \in [0, \bar{d}]$ is the dividend rate chosen by the firm. The objective is

$$V(t, x) = \sup_d \mathbb{E} \left[\int_t^T d_s ds + G(X_T) \right],$$

where G is a terminal value of remaining cash.

For a smooth candidate value function v , the HJB equation is

$$\partial_t v(t, x) + \sup_{d \in [0, \bar{d}]} \left\{ (\mu - d)v_x(t, x) + \frac{1}{2} \sigma^2 v_{xx}(t, x) + d \right\} = 0,$$

with terminal condition

$$v(T, x) = G(x).$$

The terms depending on d are

$$-d v_x(t, x) + d = d(1 - v_x(t, x)).$$

Therefore the pointwise optimizer is of bang-bang form:

$$d^*(t, x) = \begin{cases} \bar{d}, & v_x(t, x) < 1, \\ 0, & v_x(t, x) > 1, \\ \text{any value in } [0, \bar{d}], & v_x(t, x) = 1. \end{cases}$$

The financial interpretation is natural. The quantity $v_x(t, x)$ is the marginal continuation value of one additional dollar kept inside the firm. If $v_x(t, x) > 1$, then retaining one dollar inside the firm is more valuable than paying it immediately as a dividend, so the firm pays no dividends. If $v_x(t, x) < 1$, the marginal continuation value is less than the immediate dollar paid to shareholders, so the firm pays dividends at the maximal rate.

To verify optimality, suppose a smooth function v satisfies the HJB equation and define the feedback control

$$d_s^* = d^*(s, X_s^*).$$

If the closed-loop SDE

$$dX_s^* = (\mu - d_s^*) ds + \sigma dW_s$$

is admissible and the martingale term in Itô's formula is a true martingale, then

$$\partial_s v(s, X_s^*) + (\mu - d_s^*) v_x(s, X_s^*) + \frac{1}{2} \sigma^2 v_{xx}(s, X_s^*) + d_s^* = 0.$$

Hence the verification theorem gives

$$v(t, x) = V(t, x)$$

and d^* is an optimal dividend policy.

Example 3.10 (Optimal reinsurance and risk retention). An insurer chooses a retention level $a_s \in [0, 1]$, where $a_s = 1$ means the insurer retains all insurance risk and $a_s = 0$ means the insurer transfers all risk to a reinsurer. Consider the simplified surplus dynamics

$$dX_s = (\eta + \theta a_s) ds + \sigma a_s dW_s, \quad X_t = x.$$

Here η is a baseline surplus growth rate, $\theta > 0$ is the additional expected profit from retaining more risk, and σa_s is the volatility generated by retained insurance risk.

Suppose the insurer maximizes expected utility of terminal surplus:

$$V(t, x) = \sup_a \mathbb{E}[U(X_T)],$$

where U is increasing and concave. The HJB equation is

$$\partial_t v(t, x) + \sup_{a \in [0, 1]} \left\{ (\eta + \theta a) v_x(t, x) + \frac{1}{2} \sigma^2 a^2 v_{xx}(t, x) \right\} = 0,$$

with terminal condition

$$v(T, x) = U(x).$$

Assume that the candidate value function is increasing and concave, so that

$$v_x(t, x) > 0, \quad v_{xx}(t, x) < 0.$$

For fixed (t, x) , the Hamiltonian as a function of a is

$$(\eta + \theta a) v_x(t, x) + \frac{1}{2} \sigma^2 a^2 v_{xx}(t, x).$$

The first-order condition for an interior maximizer is

$$\theta v_x(t, x) + \sigma^2 a v_{xx}(t, x) = 0.$$

Thus the candidate optimal retention level is

$$a^*(t, x) = \Pi_{[0,1]} \left(-\frac{\theta v_x(t, x)}{\sigma^2 v_{xx}(t, x)} \right),$$

where $\Pi_{[0,1]}$ denotes projection onto the interval $[0, 1]$.

Because $v_{xx} < 0$, the expression

$$-\frac{v_x}{v_{xx}}$$

is the local risk tolerance of the value function. Thus the optimal retention level increases when the insurer has greater risk tolerance and decreases when volatility σ is larger.

If the feedback control

$$a_s^* = a^*(s, X_s^*)$$

is admissible and the closed-loop state process satisfies

$$dX_s^* = (\eta + \theta a_s^*) ds + \sigma a_s^* dW_s,$$

then the feedback rule attains the supremum in the HJB equation. Therefore

$$\partial_s v(s, X_s^*) + (\eta + \theta a_s^*) v_x(s, X_s^*) + \frac{1}{2} \sigma^2 (a_s^*)^2 v_{xx}(s, X_s^*) = 0.$$

The verification theorem then implies

$$v(t, x) = V(t, x),$$

and the feedback retention policy a^* is optimal.

4 Maximum principle

This section gives a more detailed treatment of Pontryagin's stochastic maximum principle. The main goal is to explain how first-order perturbations of a control lead naturally to an adjoint BSDE, and how the resulting forward-backward system characterizes optimal controls.

4.1 Backward stochastic differential equations: the basic object

The stochastic maximum principle is naturally expressed through a backward stochastic differential equation (BSDE). A BSDE is an equation solved backward from a terminal condition. Given a terminal random variable ξ and a generator F , one seeks adapted processes (Y, Z) such that

$$Y_t = \xi + \int_t^T F(s, Y_s, Z_s) ds - \int_t^T Z_s dW_s, \quad 0 \leq t \leq T. \quad (30)$$

Equivalently,

$$-dY_t = F(t, Y_t, Z_t) dt - Z_t dW_t, \quad Y_T = \xi.$$

Here Y is the backward value process and Z is the martingale integrand. In financial language, Z often plays the role of a hedging or sensitivity process; in stochastic control, it is the noise component of the adjoint variable.

The standard square-integrable spaces are

$$\mathcal{S}^2(0, T) = \left\{ Y : Y \text{ adapted and continuous, } \mathbb{E} \left[\sup_{0 \leq t \leq T} |Y_t|^2 \right] < \infty \right\},$$

$$\mathcal{H}^2(0, T) = \left\{ Z : Z \text{ progressively measurable, } \mathbb{E} \int_0^T |Z_t|^2 dt < \infty \right\}.$$

Under the usual assumptions $\xi \in L^2(\Omega, \mathcal{F}_T)$ and F Lipschitz in (y, z) with $F(\cdot, 0, 0) \in \mathcal{H}^2(0, T)$, the BSDE (30) has a unique solution $(Y, Z) \in \mathcal{S}^2(0, T) \times \mathcal{H}^2(0, T)^d$.

The reason BSDEs appear in maximum principles is duality. Perturbing a control produces a first-order perturbation of the state. The adjoint BSDE is chosen so that an integration-by-parts identity transfers the dependence on the state perturbation into a pointwise condition on the control.

4.2 From perturbations to the adjoint BSDE

We derive the adjoint BSDE from a first-order perturbation argument.

Assume for simplicity that the control set is convex and that all coefficients are differentiable. Let $\hat{\alpha}$ be a candidate optimal control with associated state $\hat{X}$. Consider a perturbation

$$\alpha_t^\varepsilon = \hat{\alpha}_t + \varepsilon \eta_t,$$

where η is an admissible direction.

Let

$$\delta X_t = \left. \frac{d}{d\varepsilon} X_t^{\alpha^\varepsilon} \right|_{\varepsilon=0}$$

be the first variation of the state. Then δX satisfies the linearized SDE

$$d(\delta X_t) = (b_x(t)\delta X_t + b_a(t)\eta_t) dt + (\sigma_x(t)\delta X_t + \sigma_a(t)\eta_t) dW_t, \quad \delta X_0 = 0,$$

where all derivatives are evaluated along $(\hat{X}_t, \hat{\alpha}_t)$.

The first variation of the objective is

$$\delta J = \mathbb{E} \left[\int_0^T (f_x(t) \cdot \delta X_t + f_a(t) \cdot \eta_t) dt + g_x(\hat{X}_T) \cdot \delta X_T \right].$$

The terms involving δX prevent a direct characterization of optimality. The key idea is to eliminate them by introducing an adapted process (Y, Z) .

We choose (Y, Z) so that the terminal and running δX -terms can be rewritten in terms of η . Specifically, we look for Y satisfying

$$\mathbb{E} \left[g_x(\hat{X}_T) \cdot \delta X_T + \int_0^T f_x(t) \cdot \delta X_t dt \right] = \mathbb{E} \int_0^T Y_t \cdot d(\delta X_t).$$

Using the linearized dynamics of δX_t , the right-hand side produces terms involving

$$b_x(t)^\top Y_t + \sigma_x(t)^\top Z_t.$$

To cancel all dependence on δX_t , we therefore define (Y, Z) by the backward equation

$$-dY_t = (b_x(t)^\top Y_t + \sigma_x(t)^\top Z_t + f_x(t)) dt - Z_t dW_t, \quad Y_T = g_x(\hat{X}_T).$$

With this choice, all δX -terms cancel, and the first variation reduces to

$$\delta J = \mathbb{E} \int_0^T \left(f_a(t) + b_a(t)^\top Y_t + \sigma_a(t)^\top Z_t \right) \cdot \eta_t dt.$$

Defining the Hamiltonian

$$\mathbb{H}(t, x, a, y, z) = f(t, x, a) + b(t, x, a) \cdot y + \text{Tr}(\sigma(t, x, a)^\top z), \quad (31)$$

we obtain

$$\delta J = \mathbb{E} \int_0^T D_a H(t, \hat{X}_t, \hat{\alpha}_t, Y_t, Z_t) \cdot \eta_t dt.$$

Thus, optimality implies the first-order condition

$$D_a \mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, Y_t, Z_t) \cdot (a - \hat{\alpha}_t) \leq 0, \quad a \in A,$$

and, under concavity, the pointwise maximization condition

$$\hat{\alpha}_t \in \arg \max_{a \in A} \mathbb{H}(t, \hat{X}_t, a, Y_t, Z_t).$$

Remark 4.1. Note that the Hamiltonian in (31) is different from the (unoptimized) Hamiltonian in (24):

$$G(t, x, a, p, M) := b(t, x, a) \cdot p + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, a) M) + f(t, x, a) \quad (32)$$

as the trace terms differ.

4.3 Sufficient stochastic maximum principle

The adjoint equation associated with a control α and state X is

$$-dY_t = D_x \mathbb{H}(t, X_t, \alpha_t, Y_t, Z_t) dt - Z_t dW_t, \quad Y_T = D_x g(X_T). \quad (33)$$

Theorem 4.2 (Sufficient stochastic maximum principle). *Let $\hat{\alpha} \in \mathcal{A}$ and let $\hat{X}$ be the associated controlled diffusion. Suppose that there exists a solution $(\hat{Y}, \hat{Z})$ to the adjoint BSDE (33) along $(\hat{X}, \hat{\alpha})$ such that*

$$\mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) = \max_{a \in A} \mathbb{H}(t, \hat{X}_t, a, \hat{Y}_t, \hat{Z}_t), \quad 0 \leq t \leq T, \quad (34)$$

a.s., and

$$(x, a) \mapsto \mathbb{H}(t, x, a, \hat{Y}_t, \hat{Z}_t) \quad \text{is concave for each } t. \quad (35)$$

Assume also that g is concave and that the integrability conditions needed below hold. Then $\hat{\alpha}$ is optimal:

$$J(\hat{\alpha}) = \sup_{\alpha \in \mathcal{A}} J(\alpha).$$

Proof. Fix any admissible control α with state X . Write $\Delta X_t = \hat{X}_t - X_t$. By concavity of g ,

$$g(\hat{X}_T) - g(X_T) \geq D_x g(\hat{X}_T) \cdot (\hat{X}_T - X_T) = \hat{Y}_T \cdot \Delta X_T.$$

Applying Itô's product formula to $\hat{Y}_t \cdot \Delta X_t$ and using the state equations and the adjoint BSDE gives, after taking expectations and dropping martingale terms,

$$\begin{aligned} \mathbb{E}[g(\hat{X}_T) - g(X_T)] &\geq \mathbb{E} \int_0^T \left[-\Delta X_t \cdot D_x \mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) \right. \\ &\quad + \hat{Y}_t \cdot (b(t, \hat{X}_t, \hat{\alpha}_t) - b(t, X_t, \alpha_t)) \\ &\quad \left. + \text{Tr}((\sigma(t, \hat{X}_t, \hat{\alpha}_t) - \sigma(t, X_t, \alpha_t))^\top \hat{Z}_t) \right] dt. \end{aligned}$$

By the definition of $\mathbb{H}$,

$$\begin{aligned} f(t, \hat{X}_t, \hat{\alpha}_t) - f(t, X_t, \alpha_t) &= \mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) - \mathbb{H}(t, X_t, \alpha_t, \hat{Y}_t, \hat{Z}_t) \\ &\quad - \hat{Y}_t \cdot (b(t, \hat{X}_t, \hat{\alpha}_t) - b(t, X_t, \alpha_t)) \\ &\quad - \text{Tr}((\sigma(t, \hat{X}_t, \hat{\alpha}_t) - \sigma(t, X_t, \alpha_t))^\top \hat{Z}_t). \end{aligned}$$

Adding the last two displays yields

$$\begin{aligned} J(\hat{\alpha}) - J(\alpha) &\geq \mathbb{E} \int_0^T \left[H(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) - H(t, X_t, \alpha_t, \hat{Y}_t, \hat{Z}_t) \right. \\ &\quad \left. - (\hat{X}_t - X_t) \cdot D_x \mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) \right] dt. \end{aligned}$$

The concavity of $(x, a) \mapsto \mathbb{H}(t, x, a, \hat{Y}_t, \hat{Z}_t)$ and the maximum condition (34) imply

$$\mathbb{H}(t, X_t, \alpha_t, \hat{Y}_t, \hat{Z}_t) - \mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) \leq D_x \mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) \cdot (X_t - \hat{X}_t).$$

Hence the integrand in the previous inequality is nonnegative, and therefore

$$J(\hat{\alpha}) - J(\alpha) \geq 0.$$

Since α was arbitrary, $\hat{\alpha}$ is optimal. □

4.4 Connection with dynamic programming

The maximum principle and the HJB equation are two views of the same optimality structure when the Markovian value function is smooth. For the problem above, define

$$v(t, x) = \sup_{\alpha \in \mathcal{A}} \mathbb{E} \left[\int_t^T f(s, X_s^{t,x,\alpha}, \alpha_s) ds + g(X_T^{t,x,\alpha}) \right].$$

The HJB equation is

$$-\partial_t v(t, x) - \sup_{a \in A} G(t, x, a, D_x v(t, x), D_x^2 v(t, x)) = 0, \quad (36)$$

where G is defined in (32).

Theorem 4.3 (Maximum principle and dynamic programming). *Suppose $v \in C^{1,3}([0, T] \times \mathbb{R}^n) \cap C([0, T] \times \mathbb{R}^n)$ and there exists an optimal control $\hat{\alpha} \in \mathcal{A}$ with associated state process $\hat{X}$. Then*

$$G(t, \hat{X}_t, \hat{\alpha}_t, D_x v(t, \hat{X}_t), D_x^2 v(t, \hat{X}_t)) = \max_{a \in A} G(t, \hat{X}_t, a, D_x v(t, \hat{X}_t), D_x^2 v(t, \hat{X}_t)), \quad (37)$$

and the pair

$$\hat{Y}_t = D_x v(t, \hat{X}_t), \quad \hat{Z}_t = D_x^2 v(t, \hat{X}_t) \sigma(\hat{X}_t, \hat{\alpha}_t) \quad (38)$$

solves the adjoint BSDE (33).

Proof. Because $\hat{\alpha}$ is optimal, the dynamic programming principle gives

$$v(t, \hat{X}_t) = \mathbb{E} \left[\int_t^T f(s, \hat{X}_s, \hat{\alpha}_s) ds + g(\hat{X}_T) \middle| \mathcal{F}_t \right]. \quad (39)$$

Equivalently,

$$v(t, \hat{X}_t) + \int_0^t f(s, \hat{X}_s, \hat{\alpha}_s) ds$$

is a martingale. Applying Itô's formula to $v(t, \hat{X}_t)$ and identifying the finite-variation part gives

$$\partial_t v(t, \hat{X}_t) + G(t, \hat{X}_t, \hat{\alpha}_t, D_x v(t, \hat{X}_t), D_x^2 v(t, \hat{X}_t)) = 0. \quad (40)$$

Since v satisfies the HJB equation (36), comparison with (40) yields

$$G(t, \hat{X}_t, \hat{\alpha}_t, D_x v(t, \hat{X}_t), D_x^2 v(t, \hat{X}_t)) = \max_{a \in A} G(t, \hat{X}_t, a, D_x v(t, \hat{X}_t), D_x^2 v(t, \hat{X}_t)),$$

which is (37).

It remains to identify the adjoint BSDE. Define

$$\hat{Y}_t = D_x v(t, \hat{X}_t), \quad \hat{Z}_t = D_x^2 v(t, \hat{X}_t) \sigma(\hat{X}_t, \hat{\alpha}_t).$$

For fixed t , the HJB equation implies

$$\partial_t v(t, x) + G(t, x, \hat{\alpha}_t, D_x v(t, x), D_x^2 v(t, x)) \leq 0$$

for all x , while equality holds at $x = \hat{X}_t$ by (40). Hence the left-hand side has a local maximum at $x = \hat{X}_t$. Differentiating with respect to x gives

$$\partial_t D_x v(t, \hat{X}_t) + \mathcal{L}^{\hat{\alpha}_t} D_x v(t, \hat{X}_t) + D_x \mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) = 0, \quad (41)$$

where $\mathcal{L}^a$ is the diffusion generator corresponding to the control value a , applied componentwise to the vector-valued function $D_x v$.

Now apply Itô's formula to $D_x v(t, \hat{X}_t)$. We obtain

$$dD_x v(t, \hat{X}_t) = \left[\partial_t D_x v(t, \hat{X}_t) + \mathcal{L}^{\hat{\alpha}_t} D_x v(t, \hat{X}_t) \right] dt + D_x^2 v(t, \hat{X}_t) \sigma(\hat{X}_t, \hat{\alpha}_t) dW_t.$$

Using (41) and the definitions of $\hat{Y}_t$ and $\hat{Z}_t$, this becomes

$$d\hat{Y}_t = -D_x \mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) dt + \hat{Z}_t dW_t.$$

Equivalently,

$$-d\hat{Y}_t = D_x \mathbb{H}(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) dt - \hat{Z}_t dW_t.$$

Finally, the terminal condition $v(T, \cdot) = g(\cdot)$ gives

$$\hat{Y}_T = D_x v(T, \hat{X}_T) = D_x g(\hat{X}_T).$$

Thus $(\hat{Y}, \hat{Z})$ defined by (38) solves the adjoint BSDE. $\square$

4.5 Example: mean–variance portfolio selection

We now illustrate the maximum-principle method in the Black–Scholes mean–variance problem. There is one riskless asset and one risky asset:

$$dS_t^0 = rS_t^0 dt, \quad dS_t = S_t(b dt + \sigma dW_t),$$

where $b > r$ and $\sigma > 0$ are constants. If α_t denotes the amount invested in the stock, then the wealth process satisfies

$$dX_t = (rX_t + \alpha_t(b - r)) dt + \sigma \alpha_t dW_t, \quad X_0 = x. \quad (42)$$

The mean–variance problem is

$$V(m) = \inf_{\alpha \in \mathcal{A}} \{ \text{Var}(X_T) : \mathbb{E}[X_T] = m \}. \quad (43)$$

Following the Lagrange multiplier reduction, first solve the auxiliary quadratic problem

$$\tilde{V}(\lambda) = \inf_{\alpha \in \mathcal{A}} \mathbb{E}[(X_T - \lambda)^2], \quad \lambda \in \mathbb{R}. \quad (44)$$

For this minimization problem, the sign convention is the reverse of the maximization theorem above: one minimizes the Hamiltonian, or equivalently applies the maximum principle to the negative payoff.

The Hamiltonian for the auxiliary problem is

$$\mathbb{H}(x, a, y, z) = (rx + a(b - r))y + \sigma az. \quad (45)$$

The adjoint BSDE is

$$-dY_t = rY_t dt - Z_t dW_t, \quad Y_T = 2(X_T - \lambda). \quad (46)$$

For a candidate optimum $(\hat{\alpha}, \hat{X}, \hat{Y}, \hat{Z})$,

$$\mathbb{H}(x, a, \hat{Y}_t, \hat{Z}_t) = rx\hat{Y}_t + a((b - r)\hat{Y}_t + \sigma\hat{Z}_t).$$

Since the control set is $\mathbb{R}$ and the Hamiltonian is affine in a , the stationarity condition is

$$(b - r)\hat{Y}_t + \sigma\hat{Z}_t = 0. \quad (47)$$

We seek the adjoint process in the affine form

$$\hat{Y}_t = \varphi(t)\hat{X}_t + \psi(t), \quad (48)$$

with deterministic C^1 functions φ and ψ . Comparing the dynamics in (46) with Itô's formula applied to (48) gives

$$\varphi'(t)\hat{X}_t + \varphi(t)(r\hat{X}_t + \hat{\alpha}_t(b - r)) + \psi'(t) = -r(\varphi(t)\hat{X}_t + \psi(t)), \quad (49)$$

$$\varphi(t)\sigma\hat{\alpha}_t = \hat{Z}_t, \quad (50)$$

with terminal conditions

$$\varphi(T) = 2, \quad \psi(T) = -2\lambda. \quad (51)$$

Using (47), (48), and (50), the feedback control is

$$\hat{\alpha}_t = \frac{(r-b)\hat{Y}_t}{\sigma^2\varphi(t)} = \frac{(r-b)(\varphi(t)\hat{X}_t + \psi(t))}{\sigma^2\varphi(t)}. \quad (52)$$

Comparing (52) with (49) yields the ODEs

$$\varphi'(t) + \left(2r - \frac{(b-r)^2}{\sigma^2}\right)\varphi(t) = 0, \quad \varphi(T) = 2, \quad (53)$$

$$\psi'(t) + \left(r - \frac{(b-r)^2}{\sigma^2}\right)\psi(t) = 0, \quad \psi(T) = -2\lambda. \quad (54)$$

Therefore

$$\varphi(t) = 2 \exp \left[\left(2r - \frac{(b-r)^2}{\sigma^2}\right) (T-t) \right], \quad (55)$$

$$\psi_\lambda(t) = -2\lambda \exp \left[\left(r - \frac{(b-r)^2}{\sigma^2}\right) (T-t) \right]. \quad (56)$$

The optimal control for the auxiliary problem is thus

$$\hat{\alpha}^\lambda(t, x) = \frac{(r-b)(\varphi(t)x + \psi_\lambda(t))}{\sigma^2\varphi(t)} = -\frac{b-r}{\sigma^2} \left(x - \lambda e^{-r(T-t)} \right). \quad (57)$$

The last expression shows the financial meaning: the investor applies a quadratic-utility Merton strategy to the excess wealth above the deterministic target account $\lambda e^{-r(T-t)}$.

The auxiliary value is obtained by completing the square, or equivalently by applying Itô's formula to

$$\frac{1}{2}\varphi(t)X_t^2 + \psi_\lambda(t)X_t.$$

One obtains

$$\tilde{V}(\lambda) = \exp \left(-\frac{(b-r)^2}{\sigma^2} T \right) (\lambda - e^{rT} x)^2. \quad (58)$$

Finally, the constrained mean-variance problem follows from the identity

$$\mathbb{E}[(X_T - \lambda)^2] = \text{Var}(X_T) + (\mathbb{E}[X_T] - \lambda)^2. \quad (59)$$

This gives the conjugate relation

$$V(m) = \sup_{\lambda \in \mathbb{R}} \left\{ \tilde{V}(\lambda) - (m - \lambda)^2 \right\}. \quad (60)$$

Let

$$\theta = \frac{b-r}{\sigma}, \quad A = e^{-\theta^2 T}.$$

The maximizer in (60) is

$$\lambda_m = \frac{m - A e^{rT} x}{1 - A} = \frac{m - \exp \left(\left(r - \frac{(b-r)^2}{\sigma^2} \right) T \right) x}{1 - \exp \left(-\frac{(b-r)^2}{\sigma^2} T \right)}. \quad (61)$$

Hence the optimal control for the mean-constrained problem is

$$\alpha_t^* = -\frac{b-r}{\sigma^2} \left(X_t^* - \lambda_m e^{-r(T-t)} \right), \quad (62)$$

and the efficient frontier is

$$V(m) = \frac{A}{1-A} (m - xe^{rT})^2 = \frac{(m - xe^{rT})^2}{\exp(\theta^2 T) - 1}. \quad (63)$$

The special case $m = xe^{rT}$ corresponds to investing only in the riskless asset, giving zero variance.

5 Deep learning methods for stochastic control

The preceding sections introduced two complementary characterizations of stochastic control problems: the dynamic programming viewpoint, which leads to the HJB equation, and the trajectory viewpoint, which leads to forward-backward stochastic systems. Deep learning enters this picture as a flexible function approximation tool. The goal is not to change the control problem, but to approximate the objects appearing in these characterizations when classical grid-based methods become infeasible.

Consider again the finite-horizon control problem

$$dX_s = b(s, X_s, \alpha_s) ds + \sigma(s, X_s, \alpha_s) dW_s, \quad X_t = x,$$

with value function

$$V(t, x) = \sup_{\alpha \in \mathcal{A}(t)} \mathbb{E} \left[\int_t^T f(s, X_s^{t,x,\alpha}, \alpha_s) ds + g(X_T^{t,x,\alpha}) \right].$$

For a smooth value function, the HJB equation is

$$\partial_t V(t, x) + \mathcal{H}(t, x, \nabla_x V(t, x), \nabla_x^2 V(t, x)) = 0, \quad V(T, x) = g(x),$$

where the Bellman operator is

$$\mathcal{H}(t, x, p, M) = \sup_{a \in A} \left\{ b(t, x, a) \cdot p + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, a) M) + f(t, x, a) \right\}.$$

5.1 Deep PDE / PINN / DGM viewpoint

The most direct deep learning approach is to approximate the value function itself by a neural network:

$$V(t, x) \approx V_\theta(t, x).$$

The derivatives appearing in the HJB equation are then computed by automatic differentiation. Define the HJB residual

$$\mathcal{R}_\theta(t, x) = \partial_t V_\theta(t, x) + \mathcal{H}(t, x, \nabla_x V_\theta(t, x), \nabla_x^2 V_\theta(t, x)).$$

A typical PINN/DGM-type [3] loss takes the form

$$\mathcal{L}_{\text{PDE}}(\theta) = \mathbb{E}_{(t,x) \sim \rho} \left[|\mathcal{R}_\theta(t, x)|^2 \right] + \lambda_T \mathbb{E}_{x \sim \nu_T} \left[|V_\theta(T, x) - g(x)|^2 \right],$$

with an additional boundary loss if the state space is restricted to a bounded domain. Here ρ is a sampling distribution over collocation points in the time–state space.

Once V_θ is learned, the feedback control is recovered by pointwise optimization:

$$\hat{a}_\theta(t, x) \in \arg \max_{a \in A} \left\{ b(t, x, a) \cdot \nabla_x V_\theta(t, x) + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, a) \nabla_x^2 V_\theta(t, x)) + f(t, x, a) \right\}.$$

When this maximization has a closed-form solution, this gives an explicit neural feedback policy. Otherwise, one may introduce a separate actor network $a_\phi(t, x)$ and train the pair (V_θ, a_ϕ) jointly. In that case, the critic V_θ approximates the value function, while the actor a_ϕ approximates the feedback optimizer in the Hamiltonian.

The advantage of this viewpoint is that it tries to learn a global approximation of the value function on the state space. The limitation is that the method must sample the time–state domain well and, for fully nonlinear HJB equations, must compute second-order derivatives of the neural network. This can become expensive and unstable in high-dimensional problems.

5.2 Deep BSDE / FBSDE viewpoint

A different approach is to use a nonlinear Feynman–Kac or BSDE representation [4]. To keep the main idea transparent, suppose that after optimizing the Hamiltonian, the HJB equation can be written in the semilinear form

$$\partial_t V(t, x) + \mathcal{L}V(t, x) + F(t, x, V(t, x), \sigma^\top(t, x) \nabla_x V(t, x)) = 0, \quad V(T, x) = g(x),$$

where

$$\mathcal{L}\varphi(t, x) = \mu(t, x) \cdot \nabla_x \varphi(t, x) + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x) \nabla_x^2 \varphi(t, x)).$$

Let X solve the forward SDE

$$dX_s = \mu(s, X_s) ds + \sigma(s, X_s) dW_s, \quad X_0 \sim \nu_0.$$

Here ν_0 may be a point mass at a fixed initial condition x_0 , or more generally an initial distribution over states.

The nonlinear Feynman–Kac formula suggests that, along the forward process X ,

$$Y_s = V(s, X_s), \quad Z_s = \sigma^\top(s, X_s) \nabla_x V(s, X_s),$$

solve the BSDE

$$Y_s = g(X_T) + \int_s^T F(r, X_r, Y_r, Z_r) dr - \int_s^T Z_r dW_r.$$

Equivalently, in differential form,

$$dY_s = -F(s, X_s, Y_s, Z_s) ds + Z_s dW_s, \quad Y_T = g(X_T).$$

The deep BSDE method parameterizes the unknown initial value and martingale integrand by neural networks:

$$Y_0 = \eta_\theta(X_0), \quad Z_s = \zeta_{\theta'}(s, X_s).$$

Given these choices, the pair $(Y^{\theta, \theta'}, Z^{\theta'})$ is defined forward in time by

$$dY_s^{\theta, \theta'} = -F(s, X_s, Y_s^{\theta, \theta'}, \zeta_{\theta'}(s, X_s)) ds + \zeta_{\theta'}(s, X_s) dW_s, \quad Y_0^{\theta, \theta'} = \eta_\theta(X_0).$$

The parameters are trained so that the terminal condition is satisfied:

$$\mathcal{L}_{\text{BSDE}}(\theta, \theta') = \mathbb{E} \left[\left| Y_T^{\theta, \theta'} - g(X_T) \right|^2 \right].$$

Thus the BSDE is solved by a shooting method: one chooses the initial value Y_0 and the process Z , evolves the BSDE forward along simulated paths, and penalizes the mismatch at terminal time.

After training, $\eta_\theta(x)$ approximates the value function $V(0, x)$ on the support of the initial distribution ν_0 . In particular, if $\nu_0 = \delta_{x_0}$, then $\eta_\theta(x_0)$ approximates $V(0, x_0)$. The network $\zeta_{\theta'}$ approximates the gradient-related process

$$\sigma^\top(t, X_t) \nabla_x V(t, X_t).$$

If $\sigma(t, x)$ is invertible, one may recover the value-gradient along sampled trajectories by

$$\nabla_x V(t, X_t) \approx \sigma(t, X_t)^{-\top} \zeta_{\theta'}(t, X_t).$$

The feedback control can then be obtained by the same pointwise optimization in the Hamiltonian, using the learned approximation of the value-gradient.

This viewpoint differs from the deep PDE/PINN/DGM approach in an important way. Instead of learning a global function $V_\theta(t, x)$ by enforcing the HJB residual at collocation points in the full time–state space, the deep BSDE method learns the value process and gradient process along simulated trajectories. It is therefore especially natural when the HJB equation is semilinear and when one is primarily interested in the value starting from a prescribed initial state or initial distribution.

Remark 5.1. *The BSDE variables in the nonlinear Feynman–Kac representation should not be confused with the adjoint variables in the stochastic maximum principle. In the BSDE representation above, Y_t is the value process $V(t, X_t)$. In the maximum principle, the adjoint process is closer to $\nabla_x V(t, X_t)$ along the optimal trajectory. These two viewpoints are related in smooth Markovian settings, but they are not the same object.*

Remark. The BSDE variables in the nonlinear Feynman–Kac representation should not be confused with the adjoint variables in the stochastic maximum principle. In the BSDE representation above, Y_t is the value process $V(t, X_t)$. In the maximum principle, the adjoint process is closer to $\nabla_x V(t, X_t)$ along the optimal trajectory. These two viewpoints are related in smooth Markovian settings, but they are not the same object.

5.3 Comparison of the two viewpoints

Method	What is learned	Main feature
Deep PDE / PINN / DGM	A global neural approximation $V_\theta(t, x)$ of the value function	Trains by minimizing the HJB residual over sampled collocation points in time–state space. It is natural when one wants a global value function and feedback policy.
Deep BSDE / FBSDE	The value process Y_t and gradient process Z_t along simulated trajectories	Trains by matching the terminal condition $Y_T = g(X_T)$. It avoids spatial grids and often scales better in high dimensions, especially for semilinear HJB equations or problems with a fixed initial distribution.

The main message is the following. Deep PDE methods approximate the value function directly and enforce the HJB equation pointwise. Deep BSDE methods approximate the value and gradient processes along simulated paths and enforce the terminal condition. Therefore, BSDE methods are often more attractive in high-dimensional problems, while PDE/PINN/DGM methods are more natural when a global approximation of $V(t, x)$ over a prescribed state domain is required.

6 Optimal stopping

In the preceding sections, the decision-maker continuously chooses a control process. In an optimal stopping problem, the decision is different: the agent observes a stochastic process over time and chooses *when to stop*. Once stopping occurs, the problem ends and the agent receives a terminal reward.

Optimal stopping problems arise naturally in many applications:

- in finance, deciding when to exercise an American option;
- in statistics, deciding when enough evidence has been collected to accept or reject a hypothesis;
- in operations, deciding when to abandon a project, replace a machine, or launch an intervention.

The key feature is the trade-off between stopping immediately and waiting for more information. Waiting may improve the future payoff, but it also exposes the decision-maker to delay costs, discounting, or unfavorable state movements.

6.1 Infinite-horizon optimal stopping

We begin with the infinite-horizon case because it highlights the core structure most clearly. Let $X = (X_t)_{t \geq 0}$ be a one-dimensional diffusion satisfying

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t, \quad X_0 = x, \quad (64)$$

where $\sigma(x) > 0$. Let $\rho > 0$ be a discount rate and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be the reward received upon stopping.

The decision-maker chooses a stopping time τ and receives the discounted payoff

$$e^{-\rho\tau}g(X_\tau).$$

The value function is

$$V(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x [e^{-\rho\tau}g(X_\tau)], \quad (65)$$

where $\mathcal{T}$ is the set of admissible stopping times and $\mathbb{E}_x$ denotes expectation conditional on $X_0 = x$.

The decision at state x is simple in principle:

- stop immediately and receive $g(x)$;
- continue, in which case the value is determined by the future evolution of X .

Therefore the value function must satisfy the inequality

$$V(x) \geq g(x). \quad (66)$$

The region where equality holds is the stopping region.

Definition 6.1 (Stopping and continuation regions). *The stopping region is*

$$\mathcal{S} = \{x : V(x) = g(x)\}.$$

The continuation region is

$$\mathcal{C} = \{x : V(x) > g(x)\}.$$

Heuristic derivation of the HJB equation. When $X_t \in \mathcal{S}$, it is optimal to stop. When $X_t \in \mathcal{C}$, it is optimal to continue.

Let $\mathcal{L}$ be the infinitesimal generator of X :

$$\mathcal{L}\varphi(x) = b(x)\varphi'(x) + \frac{1}{2}\sigma^2(x)\varphi''(x).$$

Inside the continuation region, the agent does not stop, so the value should evolve like the discounted value of the diffusion. Heuristically, the dynamic programming principle over a short time interval $[0, h]$ gives

$$V(x) = \mathbb{E}_x [e^{-\rho h}V(X_h)], \quad x \in \mathcal{C}.$$

Using Itô's formula and expanding for small h , we obtain

$$0 = \mathcal{L}V(x) - \rho V(x), \quad x \in \mathcal{C}.$$

In the stopping region, the value equals the reward:

$$V(x) = g(x), \quad x \in \mathcal{S}.$$

Combining these two conditions gives the free-boundary formulation

$$\begin{cases} \mathcal{L}V(x) - \rho V(x) = 0, & x \in \mathcal{C}, \\ V(x) = g(x), & x \in \mathcal{S}, \\ V(x) \geq g(x), & x \in \mathbb{R}. \end{cases} \quad (67)$$

Equivalently, the value function satisfies the variational inequality

$$\max \{ \mathcal{L}V(x) - \rho V(x), g(x) - V(x) \} = 0. \quad (68)$$

This equation has a natural interpretation. Since $V \geq g$, we have $g - V \leq 0$. In the continuation region, $g(x) - V(x) < 0$, so the maximum in (68) can equal zero only if

$$\mathcal{L}V(x) - \rho V(x) = 0.$$

In the stopping region, $V(x) = g(x)$, so the obstacle constraint is binding.

6.2 Free boundary and smooth pasting

In many one-dimensional examples, the stopping region is separated from the continuation region by a threshold. Suppose, for instance, that it is optimal to stop when X_t falls below a boundary a :

$$\mathcal{S} = (-\infty, a], \quad \mathcal{C} = (a, \infty).$$

Then the optimal stopping time is expected to have the form

$$\tau^* = \inf\{t \geq 0 : X_t \leq a\}. \quad (69)$$

The unknown number a is called a *free boundary*. It is not given in advance; it is part of the solution. To determine it, one solves the ODE in the continuation region and imposes boundary conditions at a .

The first condition is *value matching*:

$$V(a) = g(a). \quad (70)$$

This simply says that at the boundary, the agent is indifferent between stopping and continuing.

Under sufficient regularity and nondegeneracy, one also expects the *smooth-pasting* condition

$$V'(a) = g'(a). \quad (71)$$

This condition says that the value function touches the payoff smoothly at the stopping boundary. If the first derivatives did not match, one could usually improve the stopping boundary slightly, contradicting optimality.

Remark 6.2 (Steps to solve optimal stopping problems). *A typical one-dimensional optimal stopping calculation follows these steps:*

1. Guess the form of the stopping region, for example $x \leq a$ or $x \geq b$.
2. Solve the linear ODE

$$\mathcal{L}V - \rho V = 0$$

in the continuation region.

3. *Impose value matching and smooth pasting at the unknown boundary.*
4. *Use growth or boundedness conditions to remove inadmissible solutions.*
5. *Verify that the candidate satisfies the variational inequality.*

The resulting optimal rule is often a threshold rule: continue on one side of the boundary and stop on the other side. This is the optimal stopping analogue of a feedback control.

6.3 Example: perpetual American put option

We now derive a classical financial example. Consider a stock price following the Black–Scholes model

$$dS_t = (r - \delta)S_t dt + \sigma S_t dW_t, \quad S_0 = s > 0,$$

where $r > 0$ is the risk-free rate, $\delta \geq 0$ is the dividend yield, and $\sigma > 0$. A perpetual American put option with strike K gives the holder the right to sell the stock for K at any stopping time. Its payoff upon exercise is

$$g(s) = (K - s)^+.$$

Under risk-neutral pricing, the value is

$$V(s) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_s \left[e^{-r\tau} (K - S_\tau)^+ \right]. \quad (72)$$

For a put option, exercise is attractive when the stock price is sufficiently low. Thus we guess that there exists a boundary $s^* \in (0, K)$ such that

$$\mathcal{S} = (0, s^*], \quad \mathcal{C} = (s^*, \infty).$$

The optimal exercise time is then

$$\tau^* = \inf\{t \geq 0 : S_t \leq s^*\}. \quad (73)$$

In the continuation region $s > s^*$, the value satisfies

$$\frac{1}{2}\sigma^2 s^2 V''(s) + (r - \delta)sV'(s) - rV(s) = 0. \quad (74)$$

We look for power solutions of the form $V(s) = s^\beta$. Substituting into (74) gives

$$\frac{1}{2}\sigma^2 \beta(\beta - 1) + (r - \delta)\beta - r = 0.$$

Let β_- be the negative root:

$$\beta_- = \frac{-\left(r - \delta - \frac{1}{2}\sigma^2\right) - \sqrt{\left(r - \delta - \frac{1}{2}\sigma^2\right)^2 + 2r\sigma^2}}{\sigma^2} < 0. \quad (75)$$

The positive root is ruled out because the value of a put option should not explode as $s \rightarrow \infty$. Hence, in the continuation region,

$$V(s) = As^{\beta_-}, \quad s > s^*.$$

In the stopping region,

$$V(s) = K - s, \quad 0 < s \leq s^*.$$

We determine A and s^* using value matching and smooth pasting:

$$A(s^*)^{\beta_-} = K - s^*,$$

and

$$A\beta_-(s^*)^{\beta_- - 1} = -1.$$

Dividing the second equation by the first gives

$$\frac{\beta_-}{s^*} = -\frac{1}{K - s^*}.$$

Therefore

$$\beta_-(K - s^*) = -s^*,$$

which yields

$$s^* = \frac{\beta_-}{\beta_- - 1}K. \quad (76)$$

Since $\beta_- < 0$, we have $0 < s^* < K$.

Using value matching, we obtain

$$A = (K - s^*)(s^*)^{-\beta_-}.$$

Thus the perpetual American put value is

$$V(s) = \begin{cases} K - s, & 0 < s \leq s^*, \\ (K - s^*) \left(\frac{s}{s^*}\right)^{\beta_-}, & s > s^*. \end{cases} \quad (77)$$

Remark 6.3 (Financial interpretation). *The boundary s^* is the critical stock price below which immediate exercise is optimal. If $s > s^*$, the option holder waits because the time value of the option exceeds the immediate exercise payoff. If $s \leq s^*$, the stock price is sufficiently low that exercising the option dominates continuation.*

Remark 6.4 (American call option). *For a non-dividend-paying stock, it is never optimal to exercise a perpetual American call early. The reason is that waiting preserves upside exposure while exercise sacrifices the time value of money. With dividends, however, early exercise may become optimal because holding the option does not collect dividends.*

6.4 Example: sequential hypothesis testing

Optimal stopping also appears in statistical decision-making. We now discuss a classical hypothesis-testing example. This example is based on [5].

Suppose we observe a process whose drift Θ is unknown. There are two hypotheses:

$$H_0 : \Theta = \mu_0, \quad H_1 : \Theta = \mu_1,$$

where $\mu_1 > \mu_0$ and $\sigma > 0$ are known. The goal is to decide whether the true hypothesis is H_0 or H_1 , based on sequential observations of Y :

$$dY_t = \Theta dt + \sigma dW_t.$$

Let

$$\Pi_t = \mathbb{P}(\Theta = \mu_1 \mid \mathcal{F}_t^Y)$$

be the posterior probability of H_1 given observations up to time t . The posterior process summarizes all relevant information, so the optimal stopping problem can be written in terms of Π_t alone.

We first derive the posterior dynamics. The conditional drift of Y given the observation filtration is

$$\hat{\Theta}_t = \mathbb{E}[\Theta \mid \mathcal{F}_t^Y] = \mu_0(1 - \Pi_t) + \mu_1\Pi_t.$$

Define the innovation process by

$$d\widetilde{W}_t = \frac{1}{\sigma} (dY_t - \hat{\Theta}_t dt).$$

Then $\widetilde{W}$ is a Brownian motion with respect to the observation filtration $\mathbb{F}^Y$. The two-state filtering equation gives

$$d\Pi_t = \frac{\mu_1 - \mu_0}{\sigma} \Pi_t(1 - \Pi_t) d\widetilde{W}_t. \quad (78)$$

The posterior has no drift. This is natural because

$$\Pi_t = \mathbb{E}[\mathbf{1}_{\{\Theta = \mu_1\}} \mid \mathcal{F}_t^Y],$$

and hence $(\Pi_t)_{t \geq 0}$ is an $\mathbb{F}^Y$ -martingale. Intuitively, before seeing the next observation, the current posterior is already the best estimate of whether H_1 is true; new data may move the posterior up or down, but its conditional expected change is zero.

The statistician chooses both a stopping time τ and a terminal decision

$$D_\tau \in \{\mu_0, \mu_1\},$$

where $D_\tau = \mu_0$ means accepting H_0 , and $D_\tau = \mu_1$ means accepting H_1 . The terminal decision must be $\mathcal{F}_\tau^Y$ -measurable. With observation cost $c > 0$ per unit time and zero-one terminal loss, the Bayes risk is

$$\inf_{\tau, D_\tau} \mathbb{E} [c\tau + \mathbf{1}_{\{D_\tau \neq \Theta\}}]. \quad (79)$$

This is the standard formulation: the terminal loss is the indicator of making the wrong hypothesis decision.

Now fix a stopping time τ . Conditional on $\mathcal{F}_\tau^Y$, if the statistician chooses $D_\tau = \mu_1$, then the decision is wrong precisely when $\Theta = \mu_0$. Hence

$$\mathbb{E} [\mathbf{1}_{\{D_\tau \neq \Theta\}} \mid \mathcal{F}_\tau^Y] = \mathbb{P}(\Theta = \mu_0 \mid \mathcal{F}_\tau^Y) = 1 - \Pi_\tau.$$

If instead the statistician chooses $D_\tau = \mu_0$, then the decision is wrong precisely when $\Theta = \mu_1$. Hence

$$\mathbb{E} [\mathbf{1}_{\{D_\tau \neq \Theta\}} \mid \mathcal{F}_\tau^Y] = \mathbb{P}(\Theta = \mu_1 \mid \mathcal{F}_\tau^Y) = \Pi_\tau.$$

Therefore the optimal terminal decision at time τ is

$$D_\tau^* = \begin{cases} \mu_1, & \Pi_\tau \geq \frac{1}{2}, \\ \mu_0, & \Pi_\tau < \frac{1}{2}. \end{cases}$$

The minimized conditional probability of a wrong decision is

$$\min\{\Pi_\tau, 1 - \Pi_\tau\}.$$

Thus, after optimizing over the terminal decision, the original sequential testing problem (79) reduces to the optimal stopping problem

$$V(\pi) = \inf_{\tau} \mathbb{E}_{\pi} [c\tau + \min\{\Pi_\tau, 1 - \Pi_\tau\}], \quad \pi \in [0, 1]. \quad (80)$$

This is a cost-minimization optimal stopping problem. The agent trades off the value of more information against the cost of delay.

Let

$$a = \frac{\mu_1 - \mu_0}{\sigma}.$$

The generator of the posterior process is

$$\mathcal{L}\varphi(\pi) = \frac{1}{2}a^2\pi^2(1-\pi)^2\varphi''(\pi).$$

In the continuation region, the statistician keeps observing, and the value satisfies

$$\mathcal{L}V(\pi) + c = 0. \quad (81)$$

The stopping cost is

$$G(\pi) = \min\{\pi, 1 - \pi\}.$$

Since this is a minimization problem, stopping immediately is always feasible, so the value must satisfy

$$V(\pi) \leq G(\pi).$$

The variational inequality is

$$\min\{\mathcal{L}V(\pi) + c, G(\pi) - V(\pi)\} = 0. \quad (82)$$

The optimal rule has two stopping thresholds. When the posterior is very close to 0, the data strongly supports H_0 , so it is optimal to stop and choose H_0 . When the posterior is very close to 1, the data strongly supports H_1 , so it is optimal to stop and choose H_1 . Thus there exist thresholds

$$0 < \underline{\pi} < \frac{1}{2} < \bar{\pi} < 1$$

such that

$$\mathcal{S} = [0, \underline{\pi}] \cup [\bar{\pi}, 1], \quad \mathcal{C} = (\underline{\pi}, \bar{\pi}).$$

The optimal stopping time is

$$\tau^* = \inf\{t \geq 0 : \Pi_t \notin (\underline{\pi}, \bar{\pi})\}. \quad (83)$$

At this time, the terminal decision is

$$D_{\tau^*}^* = \begin{cases} \mu_0, & \Pi_{\tau^*} \leq \underline{\pi}, \\ \mu_1, & \Pi_{\tau^*} \geq \bar{\pi}. \end{cases}$$

Inside the continuation region, the ODE is

$$\frac{1}{2}a^2\pi^2(1-\pi)^2V''(\pi) + c = 0, \quad \pi \in (\underline{\pi}, \bar{\pi}). \quad (84)$$

At the lower stopping threshold, the statistician stops and chooses H_0 . The corresponding conditional probability of a wrong decision is π . Hence value matching gives

$$V(\underline{\pi}) = \underline{\pi}.$$

At the upper stopping threshold, the statistician stops and chooses H_1 . The corresponding conditional probability of a wrong decision is $1 - \pi$. Hence value matching gives

$$V(\bar{\pi}) = 1 - \bar{\pi}.$$

Smooth pasting gives

$$V'(\underline{\pi}) = 1, \quad V'(\bar{\pi}) = -1.$$

These four boundary conditions determine the value function and the two stopping thresholds.

In the symmetric case considered here, the observation cost is the same under both hypotheses and the terminal loss is the same for the two types of errors. Therefore the problem is symmetric around $\pi = 1/2$, and the two thresholds satisfy

$$\bar{\pi} = 1 - \underline{\pi}.$$

If we write $\alpha = \underline{\pi}$, then the two stopping thresholds are

$$\alpha \quad \text{and} \quad 1 - \alpha,$$

with $\alpha \in (0, 1/2)$. In this notation, value matching becomes

$$V(\alpha) = \alpha, \quad V(1 - \alpha) = \alpha,$$

and smooth pasting becomes

$$V'(\alpha) = 1, \quad V'(1 - \alpha) = -1.$$

Remark 6.5 (Interpretation). *The posterior Π_t measures the current strength of evidence for H_1 . When Π_t is close to 0, the evidence strongly supports H_0 ; when Π_t is close to 1, the evidence strongly supports H_1 . In both cases it is optimal to stop. When Π_t is near $1/2$, the evidence is ambiguous, and it is worth paying the observation cost to collect more information. Thus the continuation region is an interval around $1/2$, and the stopping region consists of two disjoint intervals near 0 and 1.*

Remark 6.6 (Connection to sequential probability ratio tests). *This two-threshold rule is the continuous-time analogue of the classical sequential probability ratio test. Instead of fixing a sample size in advance, the statistician stops once the posterior probability, or equivalently the likelihood ratio, exits a continuation band.*

6.5 Summary

Optimal stopping problems replace continuous action selection by the decision of when to stop. The value function is constrained to dominate the stopping payoff in maximization problems, or to lie below the stopping cost in minimization problems. The resulting HJB equation becomes a variational inequality with an obstacle.

For reward maximization, the generic infinite-horizon equation is

$$\max \{ \mathcal{L}V - \rho V, g - V \} = 0.$$

The continuation region is where $V > g$, and the stopping region is where $V = g$.

In one-dimensional problems, the stopping and continuation regions are often separated by free boundaries. These boundaries are determined by value matching, smooth pasting, and appropriate growth conditions. The American option example illustrates this structure in finance, while the hypothesis-testing example shows how the same mathematical framework governs sequential statistical decision-making.

7 Entropy-Regularized Control and Continuous-Time Reinforcement Learning

Classical stochastic control assumes that the model coefficients are known and optimizes directly over admissible controls. Reinforcement learning has a different starting point: the model is not known, and the agent must learn by interacting with the environment. A natural way to model this interaction is to allow randomized actions. In continuous time, this leads to an entropy-regularized relaxed control problem.

This section is based on [6] and [7].

Connection with classical reinforcement learning. Classical reinforcement learning is often presented in discrete time, where an agent observes a state, chooses an action, receives a reward, and transitions to a new state. The basic objects are the value function, the action-value or Q -function, policy evaluation, policy improvement, and temporal-difference learning. The continuous-time formulation below preserves exactly this logic, but the mathematical objects take differential forms. The Bellman recursion becomes an HJB equation, the discrete-time Q -function becomes a first-order infinitesimal advantage function, and temporal-difference errors become martingale increments. Entropy regularization plays the same role as in soft actor-critic and maximum-entropy RL: it encourages exploration and turns hard maximization over actions into a smooth Gibbs or softmax update.

7.1 Exploratory dynamics

We first recall the classical controlled diffusion

$$dX_s^a = b(s, X_s^a, a_s) ds + \sigma(s, X_s^a, a_s) dW_s, \quad X_t^a = x, \quad s \in [t, T], \quad (85)$$

where $a_s \in A \subseteq \mathbb{R}^m$. The corresponding finite-horizon payoff is

$$\mathbb{E} \left[\int_t^T r(s, X_s^a, a_s) ds + g(X_T^a) \middle| X_t^a = x \right]. \quad (86)$$

In the exploratory formulation, a feedback policy is a probability density

$$\pi(\cdot | s, x) \in \mathcal{P}(A),$$

and the action at (s, x) is sampled from $\pi(\cdot | s, x)$. At the level of the relaxed or averaged dynamics, the state evolves as

$$dX_s^\pi = \tilde{b}(s, X_s^\pi, \pi(\cdot | s, X_s^\pi)) ds + \tilde{\sigma}(s, X_s^\pi, \pi(\cdot | s, X_s^\pi)) dW_s, \quad X_t^\pi = x, \quad (87)$$

where

$$\tilde{b}(s, x, \pi) = \int_A b(s, x, a) \pi(a) da, \quad \tilde{\sigma} \tilde{\sigma}^\top(s, x, \pi) = \int_A \sigma \sigma^\top(s, x, a) \pi(a) da. \quad (88)$$

Thus randomizing actions changes both the drift and the covariance of the state process. The entropy-regularized value function of a fixed policy π is

$$J(t, x; \pi) = \mathbb{E} \left[\int_t^T \int_A (r(s, X_s^\pi, a) - \gamma \log \pi(a | s, X_s^\pi)) \pi(a | s, X_s^\pi) da ds + g(X_T^\pi) \middle| X_t^\pi = x \right], \quad (89)$$

where $\gamma > 0$ is the temperature parameter.

Remark 7.1 (Why relaxed controls are natural for RL). *In classical stochastic control, one often searches directly for an optimal feedback action $a^*(t, x)$. In reinforcement learning, however, deterministic optimization is not enough: the agent must also explore actions in order to learn their consequences. A randomized feedback law $\pi(\cdot | t, x)$ captures this exploration explicitly. Mathematically, this is a relaxed control; algorithmically, it is the continuous-time analogue of a stochastic policy in RL.*

The entropy term rewards policies that maintain nontrivial randomness. The temperature $\gamma > 0$ controls the exploration–exploitation trade-off. Large γ encourages diffuse policies, while small γ pushes the policy toward a deterministic optimizer. Thus entropy-regularized control provides a bridge between classical stochastic control and modern maximum-entropy reinforcement learning.

Define the Hamiltonian

$$H(t, x, a, p, M) = b(t, x, a) \cdot p + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, a) M) + r(t, x, a). \quad (90)$$

If $J(\cdot, \cdot; \pi)$ is smooth, then it satisfies the linear equation

$$\begin{aligned} \partial_t J(t, x; \pi) + \int_A [H(t, x, a, J_x(t, x; \pi), J_{xx}(t, x; \pi)) - \gamma \log \pi(a | t, x)] \pi(a | t, x) da &= 0, \\ J(T, x; \pi) &= g(x). \end{aligned} \quad (91)$$

7.2 Optimal exploratory HJB equation

The optimal value function is

$$J^*(t, x) = \sup_{\pi} J(t, x; \pi).$$

The exploratory HJB equation is

$$\begin{aligned} \partial_t J^*(t, x) + \sup_{\pi \in \mathcal{P}(A)} \int_A [H(t, x, a, J_x^*(t, x), J_{xx}^*(t, x)) - \gamma \log \pi(a)] \pi(a) da &= 0, \\ J^*(T, x) &= g(x). \end{aligned} \quad (92)$$

Fix (t, x) , and write

$$G(a) = H(t, x, a, J_x^*(t, x), J_{xx}^*(t, x)).$$

The pointwise optimization is

$$\sup_{\pi \in \mathcal{P}(A)} \int_A (G(a) - \gamma \log \pi(a)) \pi(a) da.$$

Introducing a Lagrange multiplier η for $\int_A \pi(a) da = 1$, the first-order condition gives

$$G(a) - \gamma(\log \pi(a) + 1) - \eta = 0.$$

Thus

$$\pi^*(a|t, x) = \frac{\exp\{G(a)/\gamma\}}{\int_A \exp\{G(a')/\gamma\} da'}.$$

Equivalently,

$$\pi^*(a|t, x) = \frac{\exp\left\{\frac{1}{\gamma} H(t, x, a, J_x^*(t, x), J_{xx}^*(t, x))\right\}}{\int_A \exp\left\{\frac{1}{\gamma} H(t, x, a', J_x^*(t, x), J_{xx}^*(t, x))\right\} da'}. \quad (93)$$

Remark 7.2 (Soft Bellman maximization). *The Gibbs policy (93) is the continuous-time analogue of the softmax policy in entropy-regularized discrete-time RL. Without entropy regularization, the HJB equation chooses an action by the hard maximization*

$$\sup_{a \in A} H(t, x, a, J_x, J_{xx}).$$

With entropy regularization, this hard maximum is replaced by the smooth log-sum-exp operator

$$\gamma \log \int_A \exp\{H(t, x, a, J_x, J_{xx})/\gamma\} da.$$

As $\gamma \downarrow 0$, the log-sum-exp converges formally to the hard maximum. Therefore the entropy-regularized HJB equation may be viewed as a smooth approximation of the classical HJB equation.

Substituting this expression into (92) gives the log-sum-exp form

$$\partial_t J^*(t, x) + \gamma \log \left[\int_A \exp \left\{ \frac{1}{\gamma} H(t, x, a, J_x^*(t, x), J_{xx}^*(t, x)) \right\} da \right] = 0, \quad J^*(T, x) = g(x). \quad (94)$$

7.3 Linear–quadratic case: Gaussian exploration

We now consider the infinite-horizon discounted LQ case. Here the discount rate $\rho > 0$ is retained because it is needed for well-posedness on the infinite horizon. Let

$$dX_t = (AX_t + Ba_t) dt + (CX_t + Da_t) dW_t, \quad X_0 = x, \quad (95)$$

and

$$r(x, a) = - \left(\frac{M}{2} x^2 + Rxa + \frac{N}{2} a^2 + Px + Qa \right), \quad N > 0. \quad (96)$$

For a randomized control density π_t , define

$$\mu_t = \int_{\mathbb{R}} a \pi_t(a) da, \quad \varsigma_t^2 = \int_{\mathbb{R}} a^2 \pi_t(a) da - \mu_t^2.$$

Then the exploratory dynamics are

$$dX_t^\pi = (AX_t^\pi + B\mu_t) dt + \sqrt{(CX_t^\pi + D\mu_t)^2 + D^2\varsigma_t^2} dW_t. \quad (97)$$

The entropy-regularized objective is

$$V(x) = \sup_{\pi} \mathbb{E} \left[\int_0^\infty e^{-\rho t} \left(\int_{\mathbb{R}} r(X_t^\pi, a) \pi_t(a) da - \gamma \int_{\mathbb{R}} \pi_t(a) \log \pi_t(a) da \right) dt \mid X_0^\pi = x \right]. \quad (98)$$

The HJB equation is

$$\rho v(x) = \sup_{\pi \in \mathcal{P}(\mathbb{R})} \int_{\mathbb{R}} \left[r(x, a) + (Ax + Ba)v'(x) + \frac{1}{2}(Cx + Da)^2 v''(x) - \gamma \log \pi(a) \right] \pi(a) da. \quad (99)$$

For fixed x , define

$$\mathcal{G}_x(a) = r(x, a) + (Ax + Ba)v'(x) + \frac{1}{2}(Cx + Da)^2 v''(x).$$

Using (96),

$$\begin{aligned} \mathcal{G}_x(a) = & -\frac{1}{2}(N - D^2 v''(x))a^2 + (CDxv''(x) + Bv'(x) - Rx - Q)a \\ & + Axv'(x) + \frac{1}{2}C^2 x^2 v''(x) - \frac{M}{2}x^2 - Px. \end{aligned} \quad (100)$$

Hence, if

$$N - D^2 v''(x) > 0,$$

the Gibbs density is Gaussian:

$$\pi^*(\cdot | x) = \mathcal{N}(\mu^*(x), s_*^2(x)), \quad (101)$$

with

$$\mu^*(x) = \frac{CDxv''(x) + Bv'(x) - Rx - Q}{N - D^2 v''(x)}, \quad s_*^2(x) = \frac{\gamma}{N - D^2 v''(x)}. \quad (102)$$

Under the quadratic ansatz

$$v(x) = \frac{1}{2}k_2 x^2 + k_1 x + k_0, \quad (103)$$

we obtain

$$\pi^*(\cdot|x) = \mathcal{N}\left(\frac{(k_2(B+CD) - R)x + k_1B - Q}{N - k_2D^2}, \frac{\gamma}{N - k_2D^2}\right). \quad (104)$$

The coefficient k_2 is determined by

$$\rho k_2 = \frac{(k_2(B+CD) - R)^2}{N - k_2D^2} + k_2(2A + C^2) - M. \quad (105)$$

Equivalently,

$$((B+CD)^2 + \rho D^2 - (2A + C^2)D^2)k_2^2 + ((2A + C^2 - \rho)N - 2R(B+CD) + MD^2)k_2 + (R^2 - MN) = 0. \quad (106)$$

Once k_2 is fixed, matching the linear and constant terms gives

$$k_1 = \frac{P(N - k_2D^2) - QR}{k_2B(B+CD) + (A - \rho)(N - k_2D^2) - BR}, \quad (107)$$

and

$$k_0 = \frac{(k_1B - Q)^2}{2\rho(N - k_2D^2)} + \frac{\gamma}{2\rho} \left(\log\left(\frac{2\pi e\gamma}{N - k_2D^2}\right) - 1 \right). \quad (108)$$

The mean in (104) is the exploitation component, while the variance is the exploration component. In particular, the exploration variance is linear in the temperature parameter γ .

Remark 7.3 (Exploitation mean and exploration variance). *The LQ example makes the exploration-exploitation decomposition explicit. The mean of the Gaussian policy coincides with the classical feedback action and therefore represents exploitation. The variance is proportional to the temperature parameter γ , and therefore represents exploration. In this sense, the optimal exploratory policy is not an arbitrary randomization around the classical control: its mean is chosen by the same Riccati structure as in classical LQ control, while its variance is determined by the entropy penalty and the local curvature of the Hamiltonian in the action variable.*

7.4 Cost and vanishing limit of exploration

Let V^{cl} denote the classical LQ value function without entropy regularization and without randomized controls. The classical optimal feedback has the same mean as the exploratory Gaussian policy:

$$a^*(x) = \frac{(k_2(B+CD) - R)x + k_1B - Q}{N - k_2D^2}. \quad (109)$$

The entropy-regularized exploratory value takes the form

$$V(x) = V^{\text{cl}}(x) + \frac{\gamma}{2\rho} \left(\log\left(\frac{2\pi e\gamma}{N - k_2D^2}\right) - 1 \right). \quad (110)$$

The exploration cost, defined as the loss in the original non-regularized reward from using the exploratory optimal policy, is

$$C_{\text{exp}}(x) = \frac{\gamma}{2\rho}. \quad (111)$$

Moreover,

$$\pi^*(\cdot|x) \Rightarrow \delta_{a^*(x)} \quad \text{as } \gamma \downarrow 0, \quad (112)$$

and

$$V(x) \rightarrow V^{\text{cl}}(x), \quad \gamma \downarrow 0. \quad (113)$$

7.5 From classical Q -functions to infinitesimal q -functions

In discrete-time RL, the Q -function of a policy π is

$$Q^\pi(x, a) = \mathbb{E} [r(x, a) + \beta V^\pi(X_1) \mid X_0 = x, A_0 = a].$$

It measures the value of taking action a now and then following π . The policy value is obtained by averaging over the policy:

$$V^\pi(x) = \int_A Q^\pi(x, a) \pi(a|x) da.$$

In continuous time, however, the time step becomes infinitesimal. If one applies action a only over a short interval $[t, t + \Delta t]$, then the resulting value differs from $J(t, x; \pi)$ only by order Δt . Therefore the ordinary Q -function collapses to the value function as $\Delta t \downarrow 0$. The nontrivial object is the first-order correction:

$$Q_{\Delta t}(t, x, a; \pi) = J(t, x; \pi) + q(t, x, a; \pi) \Delta t + o(\Delta t).$$

Thus q , rather than Q , is the continuous-time object that measures the local advantage of choosing action a .

7.6 Continuous-time q -function

We now return to the finite-horizon formulation. Fix a policy π . For $\Delta t > 0$, define $Q_{\Delta t}(t, x, a; \pi)$ as the value obtained by applying the constant action a over $[t, t + \Delta t]$ and then following π :

$$Q_{\Delta t}(t, x, a; \pi) = \mathbb{E} \left[\int_t^{t+\Delta t} r(s, X_s^a, a) ds + J(t + \Delta t, X_{t+\Delta t}^a; \pi) \mid X_t^a = x \right]. \quad (114)$$

Applying Itô's formula to $J(s, X_s^a; \pi)$ on $[t, t + \Delta t]$ gives

$$\begin{aligned} Q_{\Delta t}(t, x, a; \pi) &= J(t, x; \pi) \\ &\quad + \mathbb{E} \int_t^{t+\Delta t} [\partial_s J(s, X_s^a; \pi) + H(s, X_s^a, a, J_x, J_{xx})] ds. \end{aligned} \quad (115)$$

Therefore

$$Q_{\Delta t}(t, x, a; \pi) = J(t, x; \pi) + q(t, x, a; \pi) \Delta t + o(\Delta t), \quad (116)$$

where

$$q(t, x, a; \pi) = \partial_t J(t, x; \pi) + H(t, x, a, J_x(t, x; \pi), J_{xx}(t, x; \pi)). \quad (117)$$

Equivalently,

$$q(t, x, a; \pi) = \lim_{\Delta t \downarrow 0} \frac{Q_{\Delta t}(t, x, a; \pi) - J(t, x; \pi)}{\Delta t}. \quad (118)$$

Thus the usual Q -function collapses to J as $\Delta t \downarrow 0$, while the first-order correction q remains informative about the action.

7.7 Consistency condition and policy improvement

Combining the definition of q with (91), we get

$$\int_A [q(t, x, a; \pi) - \gamma \log \pi(a|t, x)] \pi(a|t, x) da = 0. \quad (119)$$

For deterministic policies and $\gamma = 0$, this reduces formally to

$$q(t, x, a^\pi(t, x); \pi) = 0.$$

Remark 7.4 (Continuous-time Bellman consistency). *Equation (119) is the continuous-time analogue of Bellman consistency. In discrete time, policy evaluation requires that the value function equal the expected one-step reward plus the continuation value. In continuous time, the corresponding condition says that the policy-average infinitesimal advantage, after subtracting the entropy correction, is zero. Thus q measures the deviation from local Bellman consistency for a particular action, while the policy-weighted average must vanish.*

Since the only action-dependent term in q is the Hamiltonian, the improved policy is

$$\pi^+(a|t, x) = \frac{\exp\{q(t, x, a; \pi)/\gamma\}}{\int_A \exp\{q(t, x, a'; \pi)/\gamma\} da'}. \quad (120)$$

Under the admissibility conditions of the theory,

$$J(t, x; \pi^+) \geq J(t, x; \pi). \quad (121)$$

Proposition 7.5 (Policy improvement). *Fix a policy π , and assume that $J(\cdot, \cdot; \pi) \in C^{1,2}$ and that the stochastic integrals below are true martingales. Let $q(\cdot, \cdot, \cdot; \pi)$ be defined by (117). Define the improved policy*

$$\pi^+(a|t, x) = \frac{\exp\{q(t, x, a; \pi)/\gamma\}}{\int_A \exp\{q(t, x, a'; \pi)/\gamma\} da'}.$$

Assume that π^+ is admissible. Then

$$J(t, x; \pi^+) \geq J(t, x; \pi).$$

Proof. For notational simplicity, write

$$J^\pi(t, x) = J(t, x; \pi), \quad q^\pi(t, x, a) = q(t, x, a; \pi).$$

Let X^{π^+} be the state process generated by the improved policy π^+ , starting from $X_t^{\pi^+} = x$. Applying Itô's formula to $J^\pi(s, X_s^{\pi^+})$ gives

$$\begin{aligned} dJ^\pi(s, X_s^{\pi^+}) = & \left[\partial_s J^\pi(s, X_s^{\pi^+}) + \int_A \left(b(s, X_s^{\pi^+}, a) \cdot J_x^\pi(s, X_s^{\pi^+}) \right. \right. \\ & \left. \left. + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(s, X_s^{\pi^+}, a) J_{xx}^\pi(s, X_s^{\pi^+})) \right) \pi^+(a|s, X_s^{\pi^+}) da \right] ds \\ & + \text{martingale term.} \end{aligned}$$

Using the definition

$$q^\pi(t, x, a) = \partial_t J^\pi(t, x) + H(t, x, a, J_x^\pi(t, x), J_{xx}^\pi(t, x)),$$

where

$$H(t, x, a, p, M) = b(t, x, a) \cdot p + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, x, a) M) + r(t, x, a),$$

we may rewrite the drift as

$$\int_A [q^\pi(s, X_s^{\pi^+}, a) - r(s, X_s^{\pi^+}, a)] \pi^+(a|s, X_s^{\pi^+}) da.$$

Therefore,

$$\begin{aligned} dJ^\pi(s, X_s^{\pi^+}) &= \int_A [q^\pi(s, X_s^{\pi^+}, a) - r(s, X_s^{\pi^+}, a)] \pi^+(a|s, X_s^{\pi^+}) da ds \\ &\quad + \text{martingale term.} \end{aligned}$$

Rearranging, integrating from t to T , and using $J^\pi(T, x) = g(x)$, we obtain

$$J^\pi(t, x) = \mathbb{E} \left[g(X_T^{\pi^+}) + \int_t^T \int_A (r(s, X_s^{\pi^+}, a) - q^\pi(s, X_s^{\pi^+}, a)) \pi^+(a|s, X_s^{\pi^+}) da ds \right].$$

Next, by the definition of π^+ , for each fixed (s, y) ,

$$\pi^+(\cdot|s, y) \in \arg \max_{\mu \in \mathcal{P}(A)} \int_A [q^\pi(s, y, a) - \gamma \log \mu(a)] \mu(a) da.$$

Hence,

$$\int_A [q^\pi(s, y, a) - \gamma \log \pi^+(a|s, y)] \pi^+(a|s, y) da \geq \int_A [q^\pi(s, y, a) - \gamma \log \pi(a|s, y)] \pi(a|s, y) da.$$

By the consistency condition (119), the right-hand side is zero. Thus,

$$\int_A q^\pi(s, y, a) \pi^+(a|s, y) da \geq \gamma \int_A \pi^+(a|s, y) \log \pi^+(a|s, y) da.$$

Equivalently,

$$- \int_A q^\pi(s, y, a) \pi^+(a|s, y) da \leq -\gamma \int_A \pi^+(a|s, y) \log \pi^+(a|s, y) da.$$

Substituting this inequality into the previous representation of $J^\pi(t, x)$ yields

$$\begin{aligned} J^\pi(t, x) &\leq \mathbb{E} \left[g(X_T^{\pi^+}) + \int_t^T \int_A (r(s, X_s^{\pi^+}, a) - \gamma \log \pi^+(a|s, X_s^{\pi^+})) \pi^+(a|s, X_s^{\pi^+}) da ds \right] \\ &= J(t, x; \pi^+). \end{aligned}$$

This proves the policy improvement inequality. $\square$

Remark 7.6 (Policy improvement as soft greedy update). *The update (120) is the continuous-time analogue of a greedy policy improvement step. In classical discrete-time RL, one improves a policy by choosing actions with larger $Q^\pi(x, a)$. In entropy-regularized RL, the hard greedy step is replaced by a softmax step. Here the same principle holds, with $q(t, x, a; \pi)$ replacing $Q^\pi(x, a)$. Actions with larger infinitesimal advantage receive larger probability under the improved policy.*

For the optimal policy, $q^*(t, x, a) := q(t, x, a; \pi^*)$ satisfies

$$\int_A e^{q^*(t, x, a)/\gamma} da = 1, \quad (122)$$

and hence

$$\pi^*(a|t, x) = e^{q^*(t, x, a)/\gamma}. \quad (123)$$

7.8 Martingale characterization of q-learning

Fix a policy π . The continuous-time q-function satisfies

$$q(t, x, a; \pi) = \partial_t J(t, x; \pi) + H(t, x, a, J_x(t, x; \pi), J_{xx}(t, x; \pi)).$$

Equivalently, along a trajectory generated by π , Itô's formula gives

$$dJ(t, X_t^\pi; \pi) + [r(t, X_t^\pi, a_t^\pi) - q(t, X_t^\pi, a_t^\pi; \pi)] dt = J_x(t, X_t^\pi; \pi)^\top \sigma(t, X_t^\pi, a_t^\pi) dW_t. \quad (124)$$

The right-hand side is a martingale increment. Hence the key implication is that, for the correct pair (J, q) , the compensated increment

$$dJ(t, X_t^\pi; \pi) + [r(t, X_t^\pi, a_t^\pi) - q(t, X_t^\pi, a_t^\pi; \pi)] dt \quad (125)$$

has conditional mean zero.

This is the continuous-time analogue of the discrete-time TD identity. In discrete-time policy evaluation, the correct value function satisfies

$$\mathbb{E}_\pi [r(X_k, A_k) + \beta V^\pi(X_{k+1}) - V^\pi(X_k) | X_k, A_k] = 0.$$

In continuous time, the one-step Bellman residual is replaced by the martingale increment (125). Thus learning (J, q) means choosing candidate functions so that this increment has no predictable drift.

More concretely, let J_θ and q_η be function approximators. Over a small time interval $[t, t + \Delta t]$, define the empirical continuous-time TD error

$$\delta_{t, \Delta t}^{\theta, \eta} = J_\theta(t + \Delta t, X_{t+\Delta t}) - J_\theta(t, X_t) + \int_t^{t+\Delta t} [r(s, X_s, a_s) - q_\eta(s, X_s, a_s)] ds. \quad (126)$$

If $(J_\theta, q_\eta) = (J, q)$, then

$$\mathbb{E}[\delta_{t, \Delta t}^{\theta, \eta} | \mathcal{F}_t] = 0.$$

Therefore a natural q-learning objective is to minimize the squared conditional moment violation, for example

$$\mathcal{L}_{\text{TD}}(\theta, \eta) = \mathbb{E} \left[\left| \delta_{t, \Delta t}^{\theta, \eta} \right|^2 \right], \quad (127)$$

possibly together with the terminal condition loss

$$\mathbb{E} \left[|J_\theta(T, X_T) - g(X_T)|^2 \right].$$

Thus the martingale characterization is useful because it turns the continuous-time Bellman equation into a sample-based regression condition: the learned value and q-functions should make the compensated value increments look like martingale noise rather than a process with systematic drift.

Remark 7.7 (Connection with q-learning). *In discrete-time q-learning, one adjusts the q-function so that the TD residual has mean zero. Here the same principle holds, but the residual is infinitesimal. The role of the Bellman residual is played by the drift of the compensated process in (125). Removing this drift is precisely the continuous-time analogue of fitting the Bellman equation from data.*

7.9 Temporal-difference form and connection to policy gradient

The differential identity (??) gives the continuous-time temporal-difference relation

$$dJ(t, X_t^\pi; \pi) + [r(t, X_t^\pi, a_t^\pi) - q(t, X_t^\pi, a_t^\pi; \pi)] dt = J_x(t, X_t^\pi; \pi)^\top \sigma(t, X_t^\pi, a_t^\pi) dW_t. \quad (128)$$

The right-hand side is a martingale increment. Hence the predictable part of the left-hand side must vanish under the correct value and q-functions. This is the precise continuous-time analogue of a temporal-difference error.

For a small time step Δt , (128) suggests the approximate discrete-time increment

$$J(t + \Delta t, X_{t+\Delta t}^\pi; \pi) - J(t, X_t^\pi; \pi) + \int_t^{t+\Delta t} [r(s, X_s^\pi, a_s^\pi) - q(s, X_s^\pi, a_s^\pi; \pi)] ds,$$

which has conditional mean approximately zero. Thus the continuous-time TD error can be viewed as the limit of the usual discrete-time TD error as the time step tends to zero.

We now connect this TD identity to policy improvement. If the exact Gibbs normalizer is available, the soft policy improvement step is

$$\pi^+(a|t, x) = \frac{\exp\{q(t, x, a; \pi)/\gamma\}}{\int_A \exp\{q(t, x, a'; \pi)/\gamma\} da'}.$$

In practice, however, one often restricts the policy to a tractable parametric family $\{\pi^\phi\}_{\phi \in \Phi}$. A natural approximate improvement step is then to project the Gibbs policy back to this family. Equivalently, for each (t, x) , one solves

$$\phi^+ \in \arg \min_{\phi' \in \Phi} D_{\text{KL}} \left(\pi^{\phi'}(\cdot|t, x) \left| \frac{\exp\{q(t, x, \cdot; \pi^\phi)/\gamma\}}{\int_A \exp\{q(t, x, a; \pi^\phi)/\gamma\} da} \right. \right). \quad (129)$$

Dropping the normalizing constant, which does not depend on ϕ' , the objective is

$$\int_A \left[\log \pi^{\phi'}(a|t, x) - \frac{1}{\gamma} q(t, x, a; \pi^\phi) \right] \pi^{\phi'}(a|t, x) da.$$

Using the score-function identity

$$\nabla_{\phi'} \pi^{\phi'}(a|t, x) = \pi^{\phi'}(a|t, x) \nabla_{\phi'} \log \pi^{\phi'}(a|t, x),$$

and ignoring the additive baseline term whose expectation against the score is zero, the gradient takes the form

$$\begin{aligned} & \nabla_{\phi'} \int_A \left[\log \pi^{\phi'}(a|t, x) - \frac{1}{\gamma} q(t, x, a; \pi^\phi) \right] \pi^{\phi'}(a|t, x) da \\ &= \int_A \left[\log \pi^{\phi'}(a|t, x) - \frac{1}{\gamma} q(t, x, a; \pi^\phi) \right] \nabla_{\phi'} \log \pi^{\phi'}(a|t, x) \pi^{\phi'}(a|t, x) da. \end{aligned} \quad (130)$$

Thus the policy parameters are pushed toward actions with large q-values and away from actions whose log-probability is already too large relative to their q-value.

Finally, the temporal-difference identity allows us to replace the unknown q-value by an observed increment. From (128),

$$q(t, X_t, a_t; \pi) dt = dJ(t, X_t; \pi) + r(t, X_t, a_t) dt - \text{martingale increment}.$$

In stochastic approximation, the martingale increment contributes zero mean noise. Therefore one may use

$$dJ_\theta(t, X_t) + r(t, X_t, a_t) dt$$

as a sample-based proxy for $q(t, X_t, a_t; \pi) dt$. This leads to the incremental actor update

$$\phi \leftarrow \phi + \alpha_\phi \left[dJ_\theta(t, X_t) + r(t, X_t, a_t) dt - \gamma \log \pi^\phi(a_t|t, X_t) dt \right] \nabla_\phi \log \pi^\phi(a_t|t, X_t). \quad (131)$$

Here J_θ is a critic approximation of the value function. The bracketed term is the continuous-time entropy-regularized TD signal: it combines the observed change in the critic, the instantaneous reward, and the entropy penalty. The score term $\nabla_\phi \log \pi^\phi(a_t|t, X_t)$ is the usual likelihood-ratio direction from policy-gradient methods.

Remark 7.8 (Actor–critic interpretation). *Equation (131) has the same structure as entropy-regularized actor–critic methods in discrete time. The critic J_θ estimates the value function, the TD signal estimates the local advantage of the sampled action, and the actor updates the policy parameters in the score-function direction. The main difference is that all objects are infinitesimal: rewards appear as $r dt$, value changes appear as dJ_θ , and Bellman residuals become martingale increments.*

7.10 Summary and connection to classical RL

The entropy-regularized continuous-time framework can be viewed as a stochastic-control foundation for several central ideas in reinforcement learning.

- **Policy evaluation.** For a fixed randomized policy π , the value function satisfies a linear policy-evaluation PDE. This is the continuous-time counterpart of evaluating V^π from the Bellman equation.
- **Policy improvement.** Given the value function or the infinitesimal q -function, the improved policy is obtained by a Gibbs update. This is the analogue of soft greedy improvement in maximum-entropy RL.
- **Soft optimality equation.** The optimal exploratory HJB replaces the hard maximization in the classical HJB equation by a log-sum-exp operator. As $\gamma \downarrow 0$, this soft HJB formally recovers the classical HJB equation.
- **Continuous-time q -learning.** The usual discrete-time Q -function becomes degenerate as the time step goes to zero, because taking an action for an infinitesimal time changes the value only by order Δt . The meaningful object is therefore the infinitesimal advantage q , defined as the first-order correction to $Q_{\Delta t}$.
- **Temporal-difference learning.** Bellman consistency becomes a martingale condition. The continuous-time TD error is the predictable drift of a candidate compensated value process; learning aims to remove this drift.
- **Policy gradient.** When the exact Gibbs policy is unavailable, one can restrict attention to a parametric policy family and update parameters using score-function gradients. The resulting update is the continuous-time analogue of entropy-regularized actor–critic methods.

Thus continuous-time reinforcement learning does not replace classical stochastic control. Rather, it extends stochastic control to settings where the model is unknown and the controller must learn through exploration. Entropy regularization makes this exploration mathematically tractable and connects the HJB framework to soft policy iteration, q -learning, temporal differences, and policy-gradient methods.

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